# Mortgage Performance and Insurance Protections after Natural Disasters*

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**Abstract**

Diminishing property insurance protections may weaken household financial resilience to natural disasters. We study the effect of California wildfires on mortgage markets and examine insurance protections using novel data on property damages, mortgage performance, and home insurance. Credit risks after natural disasters are greater than previously found. Fire damage increases severe delinquency by 3.6 percentage points and prepayment by *16 percentage points*. We also find that insurance settlements are below coverage limits—which proxy for replacement costs—for 33 percent of damaged homes. These reimbursement gaps are associated with household equity losses and lower property tax revenue from delayed rebuilding.

Keywords: wildfires, mortgage, prepayment risk, disaster risk

JEL Codes: G21; G51; Q54

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# 1 Introduction

The frequency and severity of natural disasters in the United States have increased substantially, costing over $120 billion in damages per year between 2020 and 2024 (NOAA, 2024). Resulting property damages represent large negative wealth shocks for homeowners. Such shocks can decrease property values, delay mortgage payments, restrict borrowers' access to future credit, and increase lenders' exposure to default risk. Insurance is the primary safeguard to mitigate these risks as it is supposed to fund restoration of damaged property, protect mortgage obligations, and provide liquidity for expenses such as temporary housing.[1] However, insurers in high risk areas have reduced coverage citing the rising costs to insure natural disasters.[2] Diminishing insurance protections may weaken households' financial resilience after natural disasters, which can delay rebuilding of damaged property, lower local tax revenue, and increase credit risk for lenders.

This paper studies the effect of natural disasters on mortgage markets, quantifies the extent of existing insurance safeguards, and assesses the resulting financial impact on households, lenders, and local governments. Specifically, we evaluate mortgage performance after California wildfires by constructing a novel database of property-level damage inspections merged with mortgages and examine insurance protections using administrative home insurance policy and claims data. Analysis of California wildfires provides an informative empirical setting to study the economic effects of natural disasters. Our sample captures a large increase in wildfires since 2017 that matches a sharp rise in physical risk damages from all perils nationwide.[3] Moreover, treatment status is clearly defined because our data precisely measure wildfire

[1] Crowdfunding and social networks are another source of emergency funding, although these funds tend to accrue to the most well-off (Cookson et al., 2023).

[2] Several insurers in California, which represent 55 percent of the market share in areas with high fire risk, have stopped writing or capped new policies (Dixon et al., 2018; Kaufman, 2021; Boomhower et al., 2024). Low quality insurers with high insolvency rates and low operational diversification have gained market share in Florida because traditional insurers have renewed fewer existing policies (Sastry et al., 2023).

[3] 13 of California's 20 most destructive fires (by number of structures burned) have occurred since 2017. For all perils, annual damages from billion-dollar disasters across the country total $160 billion per year since 2017, which is nearly four times the annual damages over the prior 37 years.

damage for each property and the distribution of damages is bimodal (most properties inside a burn perimeter are either undamaged or completely destroyed).

The primary contribution of this study is to show that natural disasters present greater credit risks to mortgage markets than previously quantified. We estimate a larger increase in both mortgage delinquency and prepayment rates for damaged properties than existing research, which finds that different perils (e.g., hurricanes, floods, tornadoes) and events (e.g., Hurricanes Harvey and Katrina) lead to small and short-lived increases in delinquency, default, and foreclosure because of insurance, government aid, and hazard mitigation efforts.[4] We also find that relatively few damaged properties are rebuilt, leading to reductions in local property-tax revenue that persist for two years after a fire. Finally, insurance policies and claims data suggest a notable share of claims have settlements below estimated replacement costs, potentially leaving households that sustain wildfire damage with large financial losses.

Our estimates of the prepayment and delinquency risks of natural disasters are larger than prior literature due to precise measurement of damages, which mitigates attenuation bias, and due to wildfires having a higher prevalence of total property loss. As these reduced-form parameter estimates include existing protections from insurance and government aid, risks for mortgage markets may increase if disasters worsen and safeguards weaken. For example, in California, greater wildfire risk and a rise in insurer-initiated policy nonrenewals have shifted households toward more expensive insurance with basic fire coverage through Fair Access to Insurance Requirements (FAIR) plans, the state-sponsored insurance of last resort.

The precise measurement of the data in this study is unique and allows us to disentangle the overall effects of fire exposure into the treatment effect of property damage and the spillover effects of being located in a burn area. The former effect would measure the impact of a large wealth shock to households from the loss of their property while the latter would measure the impact of more indirect channels such as a loss of local amenities or an information shock to their property's risks after a near miss. In our sample, 55 percent of

[4]See Gallagher and Hartley (2017); Kousky et al. (2020); Ratcliffe et al. (2020); Du and Zhao (2020); Issler et al. (2023); Billings et al. (2022); Panjwani (2022) and An et al. (2022, 2023).

properties and 58 percent of mortgages within the perimeter remain undamaged. We show that the property damage effects differ greatly from the spillover effects. Therefore, using burn perimeters as the sole treatment definition would introduce substantial measurement error and attenuate estimates of the effect of property damage.

Decomposing the overall effect of disaster exposure is necessary for nuanced policy discussion. Large property damage effects would suggest benefits from policies focusing on property adaptation, insurance protections, improved disaster risk modeling, and targeted assistance to damaged properties while large spillover effects would suggest benefits from policies that incentivize people to move out of high risk areas. Our findings, combined with recent studies that show the effectiveness of property adaptations and investments, suggest an important role for ex-ante disaster mitigation in reducing the physical damages from wildfires and the consequent risks to households, lenders, and local governments (Baylis and Boomhower, 2026; Fried, 2022).

For our empirical analysis, we construct a novel database linking property-level fire damage to mortgage performance and estimate the effect of 81 wildfires in California on mortgage borrowers' likelihood of delinquency (90 days or more past due) and prepayment (full balance repayment ahead of schedule). We combine wildfire perimeters, property-level damage inspection reports for wildfires that burn at least 1,000 acres and damage at least one structure, and mortgage performance data from 34 large banks between 2013 and 2020. Using a difference-in-differences (DiD) design with multiple treatments, we separately identify the impact of wildfires on both damaged and undamaged properties within fire perimeters relative to our control group of properties located 1 to 2 miles outside of the fire perimeters. We estimate an event study specification, which provides evidence in support of the identifying assumption of parallel trends and shows the impact on mortgage repayment in the 24 months after a wildfire.

We then use anonymized administrative policy and claims data for fire-related insurance coverage in California to impute payments received by destroyed properties and the corre-

sponding coverage limits that estimate ex-ante property replacement costs. We measure the reimbursement gap between incurred losses, which measures claims received, and coverage limits, which represent estimated replacement costs, for fully destroyed properties. Reimbursement gaps lead to financial losses for households compared to the counterfactual of receiving the full coverage limit. Last, we use building permits and tax assessment data to track rebuilding and public finance outcomes in the years after a fire.

Our analysis yields four key findings. First, we find that wildfires lead to large increases in the delinquency rates for properties that are damaged. Delinquency rates increase by as much as 3.6 percentage points, substantially higher than the average delinquency rates of 1.1 percent before the fire. In contrast, the delinquency rates for undamaged properties within the fire perimeter remain statistically unchanged after the fire. Different effects of wildfires on damaged and undamaged properties within a fire perimeter highlight the need to precisely measure treatment groups exposed to a wildfire using property-level damages instead of commonly used fire perimeters.

Second, we estimate that wildfires substantially increase prepayment for damaged properties by as much as 16 percentage points. Such a large increase compared to prepayment rates of 1.2 percent before the fire can pose risks to lenders, servicers, and mortgage-backed securities (MBS) investors. For lenders, prepayment results in lower interest revenue for mortgages in their portfolio, and originating a new mortgage becomes less attractive when interest rates are low. MBS investors face prepayment risks and changes to the composition of the securities pools.[5] Kahn et al. (2024) show that wildfire risks affect the whole securitization chain from mortgages to MBS cash flows and while wildfires can generate losses for MBS deals, wildfire exposure can be an important aspect of diversification for investors. Our findings rule out the traditional prepayment mechanisms of a property sale or mortgage refinancing, neither of which increase for damaged properties post-fire. A large increase in

[5]Prepayments affect security cash flow and price. Mortgage default and prepayment both lead to prepayment risks for agency MBS (backed by Government Sponsored Enterprises), while non-agency MBS face additional credit risks from defaults (Fuster et al., 2025).

prepayment, that is not driven by sales or refinances, suggests that many affected households use savings, insurance settlements, or other funds to prepay their mortgage.

Third, we find that 33 percent of insurance claims on destroyed properties face reimbursement gaps, meaning the settlement received is lower than the coverage limit, which for almost all of our sample is the insurer's ex-ante estimate of the structure's replacement cost value for a destroyed property. Back-of-the-envelope calculations suggest that a household's gross equity is about $151,650 to $233,457 lower than the counterfactual of a full insurance payment.

Fourth, based on building permits data, we confirm that households affected by a wildfire rebuild infrequently. Only 10.8 percent of owners with a damaged property file a new construction permit in California in the two years following a fire. Moreover, damaged properties transact at a higher rate in the four years following a fire, but sale prices decrease by 54 to 74 percent from before the fire, roughly equivalent to the structure share of property values in California. We also observe an uptick in new construction loans taken out *after* the damaged property is sold. Based on these findings, we infer that some homeowners pay off the mortgage, sell the property as a land sale with a destroyed structure, and the new owner borrows a construction loan to rebuild the property. The delay in rebuilding has large public finance implications as property tax revenue from damaged homes decreases by 51.2 to 56.9 percent in the three years following a fire.

Our results show that wildfires pose a greater risk to mortgage markets than found by previous studies. After large wildfires, McConnell et al. (2021) find no statistical difference in mortgage repayment, while Issler et al. (2023) find no change in mortgage default and small decreases in prepayment. An et al. (2023) find small increases in delinquency rates for mortgages, credit cards, and personal loans. We estimate that wildfires lead to a much larger increase in mortgage delinquency and prepayment. Our results differ primarily because our data provide a more precise measurement of property-level damages than these studies' use of wildfire burn perimeters.

When we conduct our analysis without separating damaged and undamaged properties inside the burn perimeter, similar to Issler et al. (2023), delinquency results are similar and prepayment results remain larger. We attribute these remaining differences to our sample being more representative of severe fires occurring after 2017 and of mortgages at large banks.[6] Through this paper's focus on mortgage delinquency and prepayment, we identify the immediate frictions that households face when determining whether to rebuild or walk away from a damaged property, and therefore complement the other studies' focus on long-run migration, home prices, and household spending patterns.

This paper is also related to the broader literature on the economic effects of natural disasters. Studies of the impact of natural disasters on household welfare consider home values, migration, and labor market outcomes. While analyses of the impact of rising sea levels on house prices has yet to reach a consensus, several papers find negative but temporary effects of hurricanes and floods on home prices (Atreya et al., 2013; Ortega and Taṣpinar, 2018; Gibson and Mullins, 2020; Fang et al., 2023; Addoum et al., 2024).[7] Similarly, large hurricanes have a small and transitory negative impact on affected individuals' employment, income, and liquidity (Farrell and Greig, 2018; Deryugina et al., 2018; Groen et al., 2020). However, several studies find that large disasters lead to higher out-migration (Deryugina et al., 2018; Boustan et al., 2020; McConnell et al., 2021).

Results from this study on wildfires generalize to the literature on credit effects of natural disasters in two ways. First, we expect the sensitivity of mortgage performance to property damage to be nonlinear. As wildfires lead to a higher likelihood of total loss, estimates from this study inform us about the tail risks. Moreover, large disasters are becoming more severe and frequent, and we would expect exposure to total losses to increase. In those cases, our

[6]Our sample keeps fires with some property damage only, while Issler et al. (2023) includes all wildfires regardless of whether they resulted in property damage.

[7]See Canals-Cerdá and Roman (2021) and Bakkensen et al. (2025) for comprehensive summaries. Several studies find a house price penalty in areas with a high sea-level rise (Bernstein et al., 2019; Baldauf et al., 2020; Keys and Mulder, 2020), while other studies do not (Atreya and Czajkowski, 2019; Murfin and Spiegel, 2020; Hino and Burke, 2021). Heterogeneous climate beliefs may drive differences in the estimated effects of climate risks on home prices and mortgage markets (Bakkensen and Barrage, 2022; Bakkensen et al., 2025).

findings are particularly informative. Second, while we expect cases of moderate damage, from floods or hurricanes, to lead to smaller prepayment effects due to lower repair costs, the insurance frictions we identify may still apply. Total losses from wildfires provide clear payment entitlements as coverage limits are usually at or above replacement cost values and verification of a total loss is simpler than partial damage. For example, a partial loss from a hurricane includes added frictions in determining the share of damage to be allocated between homeowners insurance and flood insurance. We expect that homeowners may be even more disadvantaged where such frictions exist.

In the remainder of this paper, Section 2 describes our data on wildfire-impacted areas and subsequent loan performance. Section 3 outlines our empirical strategy, and Section 4 presents the estimation results. Section 5 discusses insurance reimbursement gaps after wildfires. Section 6 highlights the implications of our findings for various agents, and Section 7 concludes.

# 2 Data

## 2.1 Data Sources

We rely on five data sets. Wildfire burn perimeters are obtained from Monitoring Trends in Burn Severity (MTBS). Parcel-level wildfire damage reports are provided by CAL FIRE's Damage Inspection (DINS) database. Parcel locations, transactions, property taxes, and building permits are obtained from CoreLogic Solutions (CoreLogic). Finally, mortgage origination and performance history are from FR Y-14M regulatory data. All results presented in the figures and tables of this paper are derived from calculations based on these five data sets unless otherwise noted. Details on the contents and usage for each of these data sets are described below.[8]

[8]All analysis was done using the high-performance computing cluster hosted by the Center for the Advancement of Data and Research in Economics (CADRE) at the Federal Reserve Bank of Kansas City (Lougee et al., 2018).

### 2.1.1 MTBS Wildfire Perimeters

Monitoring Trends in Burn Severity (MTBS) is an interagency program conducted by the U.S. Geological Survey (USGS) and USDA Forest Service that is dedicated to mapping the spatial extent of wildfires in the United States (Eidenshink et al., 2007). MTBS delineates wildfire burn perimeters by utilizing Landsat satellite imagery to analyze changes in burned areas before and after wildfire events. This data set includes all wildfires that burned over 1,000 acres in the western United States and over 500 acres in the eastern United States from 1984 through 2020. Because we focus only on wildfires for which parcel-level damage reports are also available, our sample of fires is restricted to the subset of California wildfires occurring between 2013 and 2020 that burned at least 1,000 acres and damaged at least one structure. These 81 wildfires are plotted in Figure 1. While fires that burned under 1,000 acres are not included in our sample, the 81 fires included account for 98 percent of the structure damage caused by wildfires in California from 2013 through 2020.

### 2.1.2 CAL FIRE Damage Inspection Data

CAL FIRE's Damage Inspection (DINS) database identifies the universe of parcels that were damaged or destroyed by wildfires from 2013 through 2020 (California Department of Forestry and Fire Protection, 2020).[9] For each parcel, damage is assessed by in-person visual inspections by teams of damage inspection specialists. The final data set identifies the extent of the damage (no damage, affected, minor, major, destroyed), the assessor parcel number, the parcel location (latitude/longitude), the parcel address, and limited structure characteristics.[10] Unaffected parcels are not included for most of the fires. Among the parcels that sustained wildfire damage, 92 percent are classified as destroyed (see Figure A.4). Therefore, we do not consider the intensive margin of destruction and instead classify parcels as damaged or undamaged.

[9]Data was obtained by the authors through a public records request.

[10]See Figure A.2 for an example damage report and Figure A.3 for details on each damage category.

### 2.1.3 CoreLogic Residential Public Records

We use CoreLogic public records to identify parcel-level latitude/longitude locations, addresses, and assessor parcel numbers (CoreLogic, 2020). In addition, we also obtain information on property transactions, including transaction type (e.g., mortgage, refinance, construction loan) and sale price, and property taxes, including property assessed value and taxes paid. For each fire-month, we calculate the distance (in meters) from each parcel to the closest MTBS wildfire perimeter. We retain parcels that fall either within a wildfire perimeter or 1 to 2 miles ($\sim$ 1.6 to 3.2 kilometers) outside a wildfire perimeter in a given month. For parcels falling within a wildfire perimeter, we classify them as damaged or undamaged by merging with DINS data. Recall that DINS contains the universe of damaged parcels but generally does not include undamaged parcels. Therefore, we classify any parcel with a DINS designation of "no damage" and any parcel falling within the fire perimeter but not included in DINS as undamaged.

Figure 1 displays an example of the parcels included in our analysis and the corresponding damage status for the Camp Fire, which burned over 14,000 single-family residences in Northern California in 2018. As shown, not all properties in the burn perimeter sustain damage, even for the largest fire in our sample. Across all fires, 58 percent of our mortgage sample and 55 percent of our properties sample within burn perimeters do not sustain damage, underscoring the importance of incorporating DINS data in addition to wildfire perimeters in order to identify damaged parcels.

### 2.1.4 CoreLogic Building Permits

We use CoreLogic building permits data to identify post-disaster construction activity. The data are collected from individual municipalities' permitting offices (instead of county tax assessor offices for public records data). Since our focus is on post-disaster building activity, we focus on permits with the project type description as "new construction". We also use the broader universe of all permits, which includes remodeling, roofing, electrical work, and

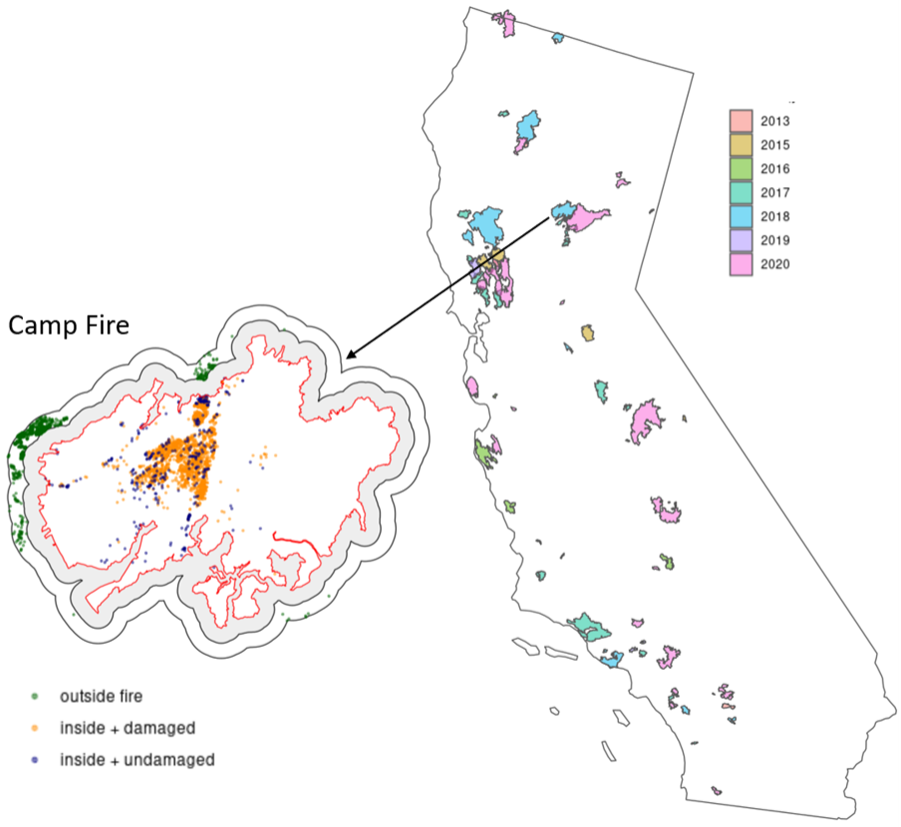



Figure 1: Donut Design of Regression Framework

Note: Figure illustrates wildfire boundaries for all wildfires between 2013 and 2020. Inset shows the wildfire boundary of the Camp Fire as well as the status of properties matched to that fire. Within the boundary, red dots denote homes that were damaged by the wildfire and green dots denote structures that were not affected by the wildfire. The gray ring is the omitted region of properties less than a mile from the wildfire boundary. The outer ring with blue dots indicates structures that are within 1 to 2 miles of the wildfire boundary.

other home improvement activities. We exclude permits that are clearly not reconstruction, such as "demolition" and "post-disaster assessment". If these permits have related rebuilding activity, they are observed as separate permits for the same property that we include. By merging this data with the CoreLogic public records data, we are able to identify which parcels were rebuilt (or in the process of being rebuilt) and which parcels did not have any new construction following a wildfire.

#### 2.1.5 Federal Reserve FR Y-14M Mortgage Performance

We obtain monthly mortgage performance history and origination characteristics for first-lien loans from FR Y-14M (Federal Reserve, 2020). These loans include both mortgages owned by the bank and mortgages that the bank continues to service. Large banks submit these data to the Federal Reserve as part of the stress testing program, with our sample including 34 institutions. These data are collected monthly starting in June 2012 and contain monthly loan performance, borrower characteristics, and property characteristics. The monthly performance history allows us to identify loans that are delinquent or have been voluntarily prepaid.[11] The borrower characteristics include credit score, debt-to-income ratio, loan-to-value (LTV) ratio, loan term, loan purpose, loan amount, and interest rate. The property characteristics include the address, which allows us to identify the property's fire damage treatment status. FR Y-14M consists of 25% of all originated loans in fire-affected tracts during our sample period, as reported in the Home Mortgage Disclosure Act (HMDA) data (Table A.1). While loan amounts are higher in the FR Y-14M, likely due to the concentration of large banks, interest rates and common measures of borrower risks (e.g. LTV, DTI) are representative of the typical mortgage in these areas.[12]

[11] When constructing our prepayment outcome, we only include voluntary prepayments as "prepaid", and not prepayments due to involuntary liquidations or servicing transfers.

[12] Since the Global Financial Crisis, non-bank lenders have originated the majority of U.S. mortgages and tend to specialize in loans that can be sold to Fannie Mae and Freddie Mac (GSEs). Banks are more likely to originate and retain non-conforming (jumbo) mortgages, which are larger than set GSE limits, on their own balance sheets as they cannot be sold to the GSEs. This distinction helps explain why average loan amounts in our Y14M sample are higher than in HMDA, which reflects the full population of lenders.

## 2.2 Data Construction

We combine the data from Section 2.1 to generate a loan-fire-month data set and a property-fire-month data set. For the loan-level data set, we match every fire event with all loans on single-family residential properties inside or within 1 to 2 miles of the wildfire boundary.[13] For each loan-fire, we then pull in the loan performance data starting 24 months before the fire through 24 months after the fire. Therefore, for each fire, we have a balanced four-year window of mortgage performance data.[14] For the property-level data set, we additionally include homes for which we do not observe a mortgage in order to track property transactions before and after the fire. We match all single-family residences inside or within 1 to 2 miles of the wildfire boundary to a damage status. As shown in Table 1, our analysis sample includes 84,413 loan-fire observations, which account for 24.8 percent of the 340,543 property-fire observations.

Table 1: Sample Construction

| Sample | Inside + Damaged | Inside + Undamaged | Outside Fire | Total |
|---|---|---|---|---|
| Single Family Houses | 23563 | 28829 | 288151 | 340543 |
| Mortgages | 5230 | 7361 | 71822 | 84413 |

Note: Table shows the decomposition of observations by treatment status. The first row shows the properties that are assigned to our treatment groups inside a fire's burn perimeter based on property damage status from inspection reports. The second row shows the set of those properties for which we observe mortgage performance. Outside fire signifies that the property or mortgage is located in the 1-2 mile ring outside a fire's burn perimeter.

While this construction is generally clean, there are a few notable complications that must be addressed. First, the same loan may be affected by multiple fires. If these fires are more than 24 months apart, then we treat them almost as separate observations (still including only one loan fixed effect). Our key assumption is that any effects of the wildfire should fade out after two years. Second, loan-fire histories may overlap (e.g., the same loan may be affected by a fire only one year after it was affected by another fire). In these cases,

[13] Over 90% of these homes are owner-occupied.

[14] Prepayment is considered a terminal state. The loan exits the sample in the month after prepayment.

we opt to keep the loans in our analysis, but we drop the pre-fire periods that overlap with the post-fire period of another fire. This does create an unbalanced panel for loans that are affected, but because of the disastrous nature of wildfires, our view is that including post-fire observations as pre-fire controls for another fire would incorrectly identify those loans as "unaffected" by a wildfire at that point in time. Addressing these data items, 98.3 percent of the remaining loans are affected by only one fire over the whole sample.

Table 2 contains borrower-level summary statistics by treatment group in the month before a wildfire. The first column presents mean borrower characteristics, loan terms, and property characteristics for all properties in our sample outside the fire zone, which represents our control group. Columns 2 and 3 reports the mean differences, relative to the control group, for damaged properties and undamaged properties inside the fire zone, respectively. Columns 4 and 5 present analogous within-fire mean differences relative the control group. Loan terms and measures of borrower risk (e.g. credit score, debt-to-income ratio, and loan-to-value ratio) are similar for mortgages that are damaged by fires to mortgages that are outside the burn perimeter.

We do note that loan amounts are larger inside the fire zone, which seems to be represented by larger differences in property characteristics across different treatment groups. Properties inside the fire perimeter are larger and located on bigger lots. Furthermore, undamaged homes inside the fire zone are more newly built. These property differences may impact the likelihood of burning – we further discuss how these differences affect the interpretation of our results in section 4.

## 2.3 Descriptive Analysis

Before proceeding with our empirical strategy, we visually summarize the impact of wildfires on prepayment and delinquency for loans falling within a wildfire perimeter in Figure 2. As shown in the top panel, our sample contains 12,591 parcels inside a wildfire perimeter at the onset of a fire, only 5,230 of which sustain damage. This once again highlights the importance

Table 2: Comparison of Summary Statistics across Treatment Groups

| | Outside Fire | Damaged | Undamaged | Damaged | Undamaged |
|---|---|---|---|---|---|
| | Mean | Mean Difference | | Mean Difference within Fire | |
| Annualized Interest Rate (%) | 4.055*** | 0.137 | -0.124** | -0.053 | -0.170*** |
| | (0.028) | (0.097) | (0.046) | (0.049) | (0.032) |
| Log(Loan Amount) (Log $) | 12.667*** | -0.178 | 0.336*** | 0.215* | 0.347*** |
| | (0.040) | (0.216) | (0.097) | (0.102) | (0.073) |
| Remaining Balance (%) | 80.414*** | 1.403* | 2.743*** | 0.699 | 1.233** |
| | (0.299) | (0.559) | (0.753) | (0.456) | (0.474) |
| Remaining Term | 258.562*** | 1.917 | 10.385** | 3.3024* | 7.344** |
| | (1.203) | (2.172) | (3.238) | (1.495) | (2.379) |
| Credit Score | 746.904*** | 0.858 | 6.743*** | 1.473 | 7.666*** |
| | (1.610) | (3.014) | (1.769) | (2.674) | (1.563) |
| LTV | 63.219*** | 2.005 | -2.709* | -3.000* | -3.490*** |
| | (0.586) | (2.879) | (1.074) | (1.266) | (0.817) |
| Debt-to-Income | 25.728*** | -2.267*** | -1.033** | -0.901** | -1.185** |
| | (0.277) | (0.495) | (0.395) | (0.314) | (0.379) |
| Delinquency Rate | 0.014*** | -0.002 | -0.003 | 0.005* | 0.001 |
| | (0.001) | (0.003) | (0.002) | (0.002) | (0.001) |
| Prepayment Rate | 0.017*** | -0.005* | -0.005* | -0.000 | 0.001 |
| | (0.001) | (0.002) | (0.002) | (0.002) | (0.002) |
| Year Built | 1973.895*** | 4.268 | 9.618*** | 3.321 | 11.511*** |
| | (1.321) | (3.557) | (1.587) | (4.042) | (1.702) |
| Building Size (Sq. Ft.) | 2126.627*** | 21.991 | 722.107*** | 467.410** | 771.830*** |
| | (65.070) | (217.752) | (163.568) | (142.456) | (153.139) |
| Lot Size (Sq. Ft.) | 25141.327*** | 77279.558** | 74086.126*** | 86264.122*** | 77011.933*** |
| | (2228.999) | (24960.132) | (20266.620) | (20019.476) | (20421.854) |
| Burn Probability | 0.002*** | 0.002* | 0.004*** | 0.003*** | 0.005*** |
| | (0.000) | (0.001) | (0.001) | (0.000) | (0.001) |

Note: Table contains loan-level summary statistics by treatment group in the month before the wildfire (event time $t = -1$). Credit score, debt-to-income (DTI), loan-to-value (LTV), and loan amount are as of origination. Interest rate varies by month for adjustable rate mortgages. Burn probability is the USFS Wildfire Risk to Communities estimated annualized probability of a wildfire burning within 270 meter cells. Delinquent is equal to one if the loan is at least three months behind in event time $t = -1$ and zero otherwise. Prepayment is equal to one if the loan is prepaid in event time $t = -1$ and zero otherwise. Columns 2 and 3 show the mean differences for damaged and undamaged properties inside any burn perimeter relative to properties in the 1 to 2 mile ring outside any burn perimeter. Columns 4 and 5 add fire fixed effects and show the within-fire mean differences for damaged and undamaged properties, relative to properties outside the burn perimeter. Stars in column 1 report significance level of whether means are statistically different from zero. Stars in columns 2 through 5 report significance level of whether means are statistically different than the corresponding group of properties outside the burn perimter.

of incorporating parcel-level damage reports into our analysis; it is not reasonable to assume that all parcels falling within a wildfire perimeter are damaged. Finally, the top panel shows the loan status 24 months post-fire.[15] The bottom panel shows that prepayments increase immediately in the month of the fire and further spike one to two months after the fire. Borrowers also become past due on their mortgage within one to two months after the fire, but the overall share of loans past due subsides six months after the fire. Two years after the fire, most of these delinquencies become current or prepay, with very few loans entering foreclosure.

# 3 Analysis

Our goal is to examine the impact of wildfire damage to one's house on delinquency and prepayment risk.[16] In order to estimate the causal effect of wildfires on mortgage repayment, we use a difference-in-differences (DiD) framework with varying treatment intensities. We use quasi-experimental variation from California wildfires, which we measure with our unique data on burn perimeters and property-level damage inspections, to disentangle the impact of fires on damaged homes from the effect on undamaged homes that are within the fire perimeters. For both treatment intensities, our control group consists of properties that are 1 to 2 miles outside of the wildfire boundary. We exclude homes in the 1-mile ring outside of the wildfire and utilize a donut design to mitigate bias from spillover effects on homes directly outside of the fire perimeter.[17]

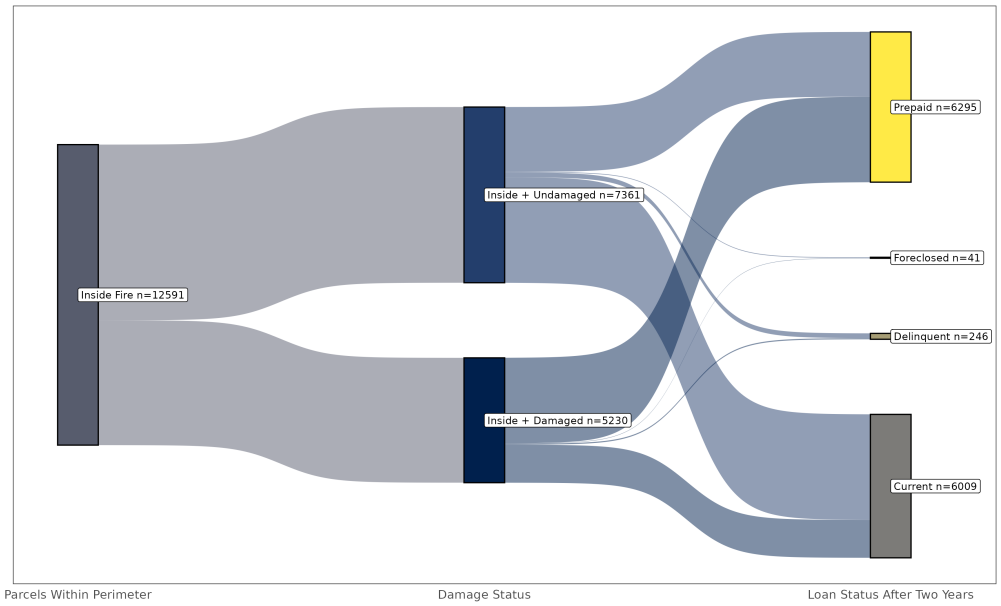
We use the following framework,

$$Y_{ift} = \beta_1 Damaged_{ift} + \beta_2 InsideFireZone_{ift} + \lambda_{ft} + \lambda_i + \epsilon_{ift},$$

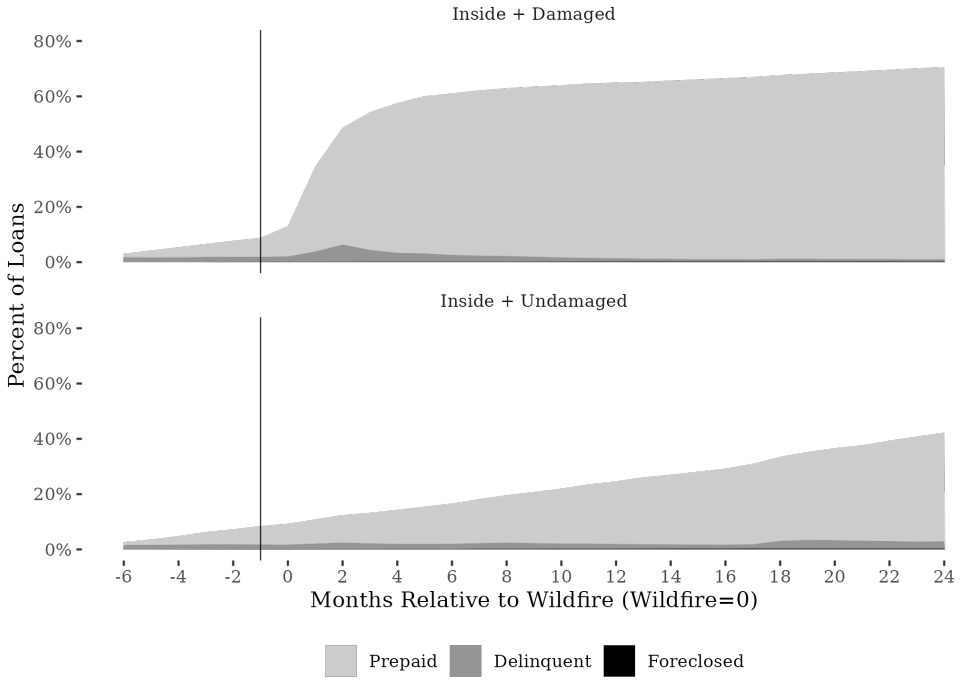
[15] Figure A.5 provides snapshots of loan performance three and six months after a fire.

[16] Throughout this paper, we use "loan" and "home" interchangeably because it is too cumbersome to write "whether loan $i$ was secured by a property that was damaged by a wildfire." We prefer to refer to homes being affected or damaged by wildfires, while loans become delinquent or are prepaid.

[17] Issler et al. (2023) discuss that homes in the 1-mile ring outside of the perimeter were often visually exposed to the fire and estimate large externalities of the wildfires on these homes.




(a) Loan Status Two Years After Fire




(b) Monthly Loan Status

Figure 2: Loan Status Flows After Wildfires

Note: Figure illustrates the decomposition of loans within wildfire boundaries based on whether or not the home was damaged by the wildfire. The top panel shows mortgage status two years post-fire (current, delinquent, foreclosed, or prepaid), and the bottom panel tracks monthly loan status.

where $Y_{ift}$ is an indicator for our outcome of interest (delinquency or prepayment). $Damaged_{ift}$ is an indicator for whether home $i$ was damaged by fire $f$ in month $t$. $InsideFireZone_{ift}$ is an indicator for whether home $i$ was not damaged but inside the perimeter of wildfire $f$ in month $t$. $\lambda_{ft}$ consists of fire-year-month fixed effects, to control for different time paths for different fires (i.e., separately capture the behavior of nondamaged loans for each fire event). Finally, $\lambda_i$ is a loan-level fixed effect.

$\beta_1$ and $\beta_2$ are our coefficients of interest. $\beta_1$ estimates the effect of having one's home damaged by a wildfire on prepayment or delinquency relative to homes 1 to 2 miles outside of the wildfire perimeter. $\beta_2$ estimates the effect of having one's home inside a wildfire perimeter (but *not* damaged) on prepayment or delinquency relative to homes 1 to 2 miles outside of the wildfire perimeter. Inside the burn perimeter, even undamaged houses may be affected by spillover effects of the wildfire (e.g., having to evacuate, dealing with damaged infrastructure, job interruptions, loss of amenities). On the other hand, homes 1 to 2 miles away from the wildfire boundary should not be affected (or at least have a substantially reduced risk of being affected) by the wildfire.

Separately identifying the direct effect of property damage ($\beta_1$) and spillover effects of being in a burn area ($\beta_2$) has important policy implications. The former mechanism measures the impact of large wealth shocks to households while the latter measures the impact of loss of local amenities or information shocks about disaster risks. In our sample, 58 percent of mortgages within the perimeter remain undamaged. Therefore, fire perimeters are insufficient proxies for property damage and may lead to inaccurate policy conclusions especially if property damage effects differ greatly from spillover effects.

This identification strategy relies on the parallel trends assumption. To provide some evidence that parallel trends hold in the pretreatment periods, we estimate an event study specification:

$$Y_{ift} = \sum_{t=-24;t\neq-1}^{t=24} \beta_{1t} Damaged_{ift} + \sum_{t=-24;t\neq-1}^{t=24} \beta_{2t} InsideFireZone_{ift} + \lambda_{ft} + \lambda_i + \epsilon_{ift}. \quad (1)$$

Several recent studies have documented the challenges of using such two-way fixed-effect regressions to identify the average treatment-on-the-treated effect, $ATT$ (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021; Borusyak et al., 2024; Sun and Abraham, 2021). These papers show that applications often use earlier-treated groups as controls for later-treated groups, which can lead to estimation bias, and may not contain never-treated groups, which can lead to underidentification.

To address these concerns, we follow a "stacked regression" design in which $\lambda_{ft}$ interacts our time fixed effect with each fire event (Cengiz et al., 2019; Baker et al., 2022; Bradt and Aldy, 2025). Therefore, these threats to identification do not apply to our within-fire comparison because, for each fire, treatment occurs simultaneously at a single date for the treatment group, and the control group remains as the never-treated group. Our results do not change if we include a vector $X_{ift}$ on the right hand side of Equation 1 to control for time-varying borrower and loan characteristics, such as current interest rate, credit score, and the share of balance remaining. The similarity in results across specifications alleviates concerns that the impact of treatment on these controls may bias estimates of $\beta_1$ and $\beta_2$ (Caetano and Callaway, 2023; Roth et al., 2023). Finally, we report standard errors clustered at the zip code level for all of our results.

# 4 Results

The top panel of Figure 3 shows that there is little evidence of pre-trends in delinquency for loans within the wildfire perimeter (whether damaged or not). We see a dramatic increase in delinquency starting three months after the fire and peaking at 3.6 percentage points within four to six months after the fire.[18] The increase then gradually declines in the six to 24 months after the fire, likely either due to loans curing (from being made whole from insurance payments) or because loans continue into default and are no longer reported as

[18]Since we define delinquency as 90 days past due, three months is the fastest a loan can become delinquent if the fire is the triggering event.

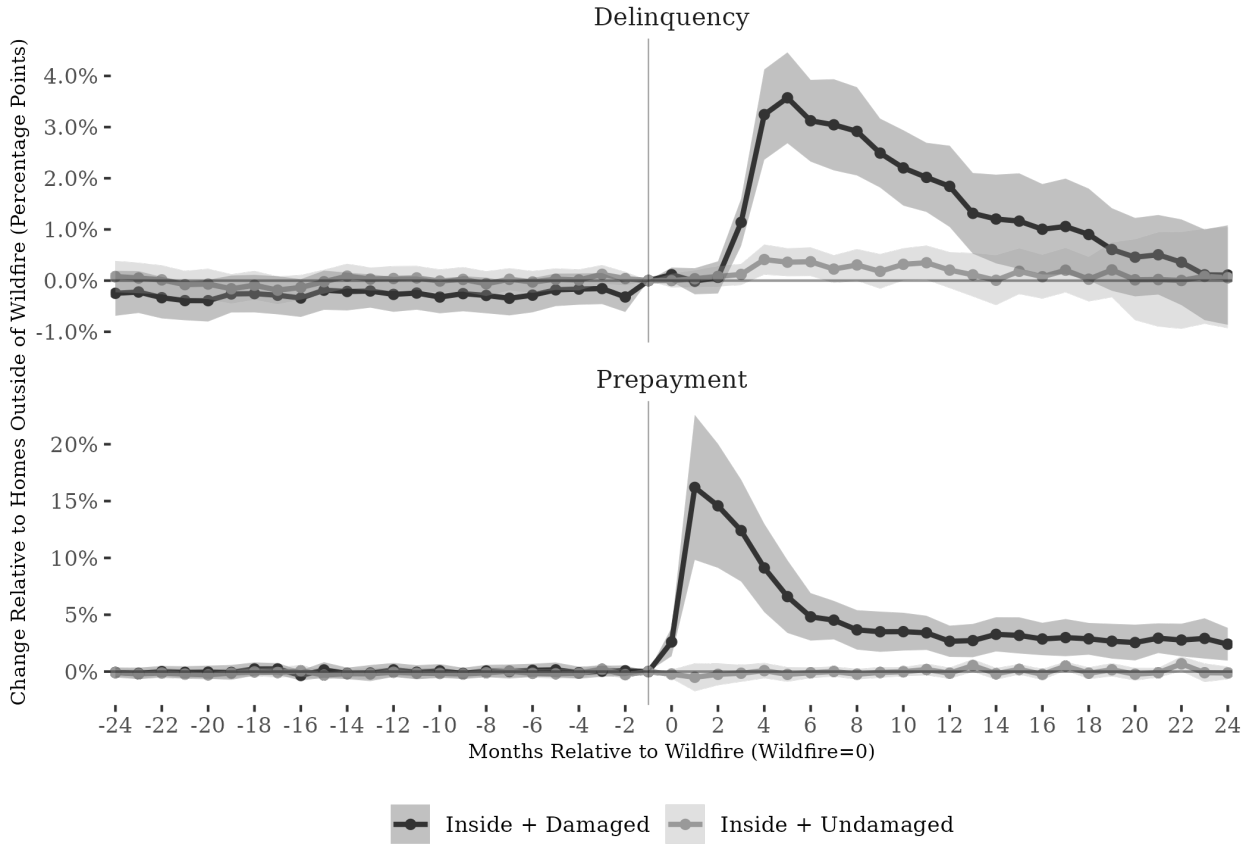



Figure 3: Event Study of Wildfire on Delinquency and Prepayment

Note: Figure illustrates the coefficients for damaged ($\beta_{1t}$) and undamaged ($\beta_{2t}$) parcels estimated from the event study specification in Equation 1. The top panel presents results for delinquency and the bottom panel presents results for prepayment. Regression controls for fire-month and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.

delinquent. For similar homes within the wildfire boundary that are not damaged by the wildfire, there is no significant change in delinquency after the fire. At best, there may be a muted 0.4 percentage point increase in delinquency four months after the fire, but this quickly fades back to be indistinguishable from zero. While there may be some minor effects of wildfire on undamaged houses, those possible effects pale in comparison with the dramatic increase in delinquency seen for damaged houses.

The 3.6 percentage point increase in delinquency is *three times* the average delinquency rate of 1.1 percent. Such a large increase in delinquency may impact banks' liquidity but provides a transitory view of mortgage performance. Therefore, we focus the remainder of

the paper on the impact of wildfires on mortgage prepayment, which is the most common terminal state for our sample of loans inside fire perimeters. Within two years after a fire, almost all of the loans inside a fire perimeter are either current or are prepaid (Figure 2). In our sample, only 41 properties are foreclosed and 246 remain delinquent after two years.

The bottom panel of Figure 3 shows that there is little evidence of pre-trends in prepayment for loans within the wildfire perimeter (whether damaged or not). Immediately after the fire, we see a dramatic increase in prepayment of about 16 percentage points; it then gradually declines (but is still elevated) for 24 months after the fire. For similar homes within the wildfire boundary that are not damaged by the wildfire, there is no significant change in prepayment after the fire. As a benchmark, the 16 percentage point increase in prepayment is *thirteen times* the average prepayment rate of 1.2 percent. Accumulating these impacts over time, we find that 68 percent of damaged properties prepay within two years after a fire. Properties that are undamaged or outside the fire perimeter prepay at a substantially lower rate of 37 percent and 33 percent, respectively. If we exclude partially damaged homes our estimated effects are even larger. Figure A.6 shows that delinquency rates are up to 4.3 percentage points higher and prepayment rates are 19.2 percentage points higher in the immediate aftermath of a fire.

## 4.1 Robustness

The lack of evidence for pre-trends suggests the parallel trends assumption likely holds. As a result, our estimates of $\beta_1$ and $\beta_2$ identify the average treatment-on-the-treated effect, $ATT$, for two treatment groups relative to homes outside of fire perimeters: damaged properties and undamaged properties inside a fire perimeter. However, this does not imply that we recover the average treatment effect, $ATE$, which requires that both burn perimeters and fire damage occur randomly. While the timing of the fire is plausibly random, one concern may be that the assignment of a property to a damaged treatment status may not be random as

household investments in property adaptations impact the likelihood of burning (Baylis and Boomhower, 2026).[19] Randomization in the setting of a natural disaster likely does not exist.

While the $ATE$ is additionally useful to understand the impacts of a treatment if it were to be expanded to areas that do not experience fires, valid identification of the $ATT$ measures the impacts on fire-affected homes. The $ATT$ is particularly informative in settings, similar to this study, where researchers are interested in the counterfactual of reducing a detrimental treatment (e.g. reducing smoking among smokers or limiting fire damage among exposed homes). For our research question, the $ATT$ measures the financial benefits of policies or mitigation efforts that would prevent fire damage in areas where fires are prevalent. We conduct several robustness checks to ensure our results are not being driven by homeowner selection into fire-affected areas, outlier fires, and misspecification of the control group.

#### 4.1.1 Selection into Fire Affected Areas

Unobserved credit risk of households who choose to live in areas and homes more likely to be damaged may be different than other households and therefore presents a threat to identification. As noted in Table 2, borrower characteristics, such as credit score, LTV, and DTI, are similar both inside and outside the burn perimeter, but homes and mortgages inside the burn perimeter tend to be larger and costlier. However, these differences are much smaller between a damaged home and an undamaged home inside a burn perimeter – that is, both treatment groups' homes have similar observable characteristics. When we estimate our model only on homes inside the burn perimeter with damaged homes serving as the treatment group and undamaged homes inside the perimeter serving as the control group. We find nearly identical results to our baseline specification (Figure A.7).

[19] Johnson et al. (2025) provides evidence that homeowners may strategically defer roof replacement in hope that a hurricane or storm claim will shift the cost to the insurer. This mechanisms relies on near-certain maintenance and replacement costs that could be deferred or offloaded to the insurer. Effective wildfire adaptations, such as defensible space, ember-resistant vents, or non-combustible siding, are elective investments instead of routine maintenance. Therefore, homeowners do not face the same incentive to strategically defer them in anticipation of an insurer-funded replacement following a fire.

Some concern remains that unobserved differences lead to higher likelihood of burning for properties within the same burn perimeter and may be correlated with credit risk. For example, homes built under modern building codes may be more fire-resistant and homeowners who buy newer homes may be different than those that buy older homes. To address this selection concern, we estimate the inverse propensity weighted treatment effect for damaged properties, relative to undamaged properties in the burn perimeter. To construct the weights, we estimate propensity of property damage as a function of the property characteristics (size, lot size, construction year, and USFS estimated burn probability) and borrower characteristics in Table 2. Since our propensity score includes construction year, it accounts for differences in fire resistance due to code changes or building age.[20] Figure A.8 shows similar results.[21] Therefore, we conclude that these selection concerns do not substantially bias our estimates.

#### 4.1.2 Exclude Camp Fire

The largest event in our analysis, the Camp Fire in 2018, damaged 2,623 homes in our mortgage sample, representing 50 percent of all damaged mortgages. After we exclude this event, Figure A.10 confirms that the pattern of findings is qualitatively similar and the increase in delinquency rates is identical. Although the increase in prepayment is smaller, the effect is economically meaningful. Prepayment rates for homes damaged by all other fires increase by 9 percentage points relative to homes outside of the fire perimeters.

#### 4.1.3 Varying Control Group

Figure A.11 shows that our results are robust to changing our control group to be homes located within 5 to 6 miles (8 to 9.6 kilometers) outside of the fire perimeter. Choosing a

[20] Construction year could be endogenous because of redevelopment incentives and may reveal something important about the homeowner's type that correlates with prepayment (Davis et al., 2021). However, many effective wildfire risk-reduction treatments are relatively cheap, such as maintaining defensible space around the house, cleaning gutters, installing ember resistant vents and vent covers, and installing spark arrestors on chimneys. These non-structural adaptations would not affect the construction year of the home or redevelopment incentives and therefore reduces this endogeneity threat.

[21] Figure A.9 shows a sufficient common support in the distribution of propensity scores by treatment group.

control group farther from the fire perimeter reduces the likelihood of spillover effects from the fire on homes outside the perimeter. While increasing distance may reduce spillovers, it may increase the chance that more distant homes are less similar to closer homes. Furthermore, increasing the perimeter increases the likelihood that perimeters from multiple fires intersect.

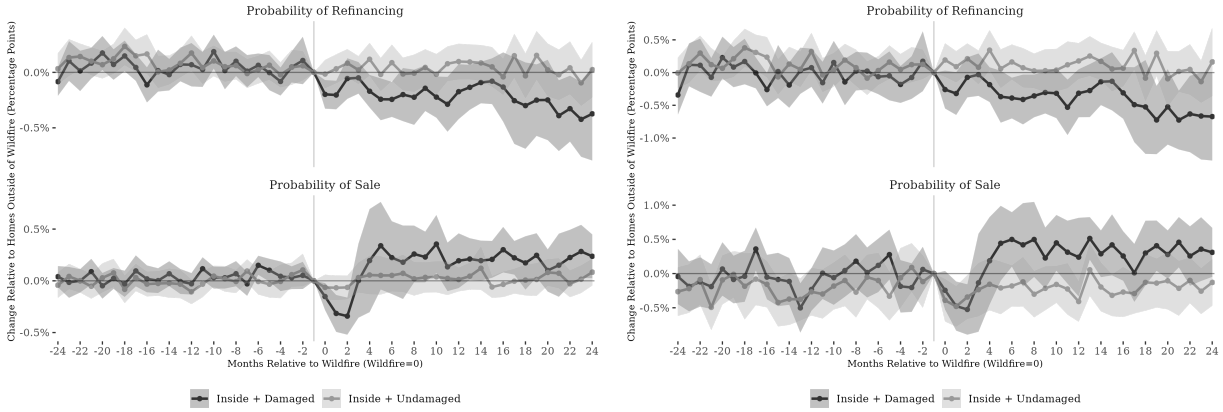
## 4.2 Home Sales and Refinance

In the aftermath of a wildfire, prepayments can potentially be driven by home sales, refinancing, or household liquidity from savings and insurance settlements.[22] We test the first two channels using observed timing of sale and refinance transactions associated with prepaid loans affected by a fire in our sample. Only 35 percent of damaged properties that prepay are sold within two years of a fire. In comparison, 47 percent of undamaged properties and 50 percent of properties outside a fire perimeter that prepay are sold within two years after a fire.

We estimate Equation 1 using indicators for whether a property was refinanced or sold in the months leading up to and after a wildfire. Figure 4 shows results from this estimation using two samples. Panel (a) uses the sample of all properties in the vicinity of a wildfire from CoreLogic public records, and Panel (b) uses the same sample of mortgages from FR Y-14M as our estimation of the impact on mortgage repayment. Results from both samples confirm that sales activity for damaged parcels does not increase until at least four months after a fire. In fact, these results highlight an immediate decline in sales for damaged properties.

Refinance activity among damaged properties does not differ from that of unaffected properties until at least 10 months after a fire, when we see refinancing among damaged

[22] Households can also receive disaster aid from the FEMA Individuals and Households Program (IHP) or the Small Business Administration (SBA) Disaster Loan program. We believe this funding is unlikely to be a key driver of prepayment because these funds provide small amounts of aid to few people. In California, from 2012 to 2019, only 350 SBA disaster loans were given out and the median IHP grant across 5,349 approved IHP grants was $1,838. IHP funds are also capped at $32,000 to $36,000 depending on the year. SBA loans are designed to cover *uninsured* losses that homeowners face, while the IHP provides money for temporary housing, housing repairs (but only to make a home safe, not to restore it to pre-disaster condition), essential needs (e.g., food, water, clothing), and funeral expenses.




(a) Properties affected by wildfire (b) FR Y-14M mortgages affected by wildfire

Figure 4: Home Sales and Refinancings Around Wildfire Events

Note: Figure illustrates the coefficients for damaged ($\beta_{1t}$) and undamaged ($\beta_{2t}$) parcels estimated from the event study specification in Equation 1, with indicators for refinance and sales, respectively, as the outcome variable. The left panel presents results for the sample of all properties in the vicinity of a fire, and the right panel presents results for the sample of mortgages we use to estimate mortgage repayment outcomes. Regression controls for fire-month and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.

properties decline.[23] An increase in sales occurring more than four months after a fire clearly does not lead to the large spike in prepayment we observe in the first three months after a fire, as shown in Figure 3. Furthermore, the estimated impact of wildfire damage on sales is significantly lower than the 16 percentage point increase in prepayment.

However, it is possible that increased sales contribute to the persistently higher prepayment rates for damaged properties after six months, as we see the rate of sales for damaged properties consistently exceeding the rate for properties that are undamaged or outside the fire zone. We track the sale of affected properties by estimating an annual event study over a eight year window around a fire. The top panel of Figure 5 shows that the sales rate for damaged homes increases by 4 to 5.7 percentage points in the four years after a fire, relative to properties outside the fire zone.[24] The bottom panel shows that the sale price for damaged homes is

[23]The long-term decline in refinance may be explained by the higher sales rate for these properties. It is less likely that these newly purchased homes would be immediately refinanced. A decline in refinancing further rejects the hypothesis that prepayments are driven by refinancing.

[24]These annual event study coefficients are consistent in magnitude with the monthly coefficients in Figure 4. The monthly coefficients show that the uptick in sales in the first year is driven by an increase in sales starting 5 months after a fire.

significantly lower in the years following a fire. Sales prices in the four years after the fire for damaged homes are 54 to 74 percent lower than the year before the fire.[25] This drop in price is similar to the average structure value as a share of property value of 57 percent for our sample of zip codes in the year before the fire (Davis et al., 2021). These results indicate that the damaged property transactions reflect land sales where homeowners sell their property without rebuilding.

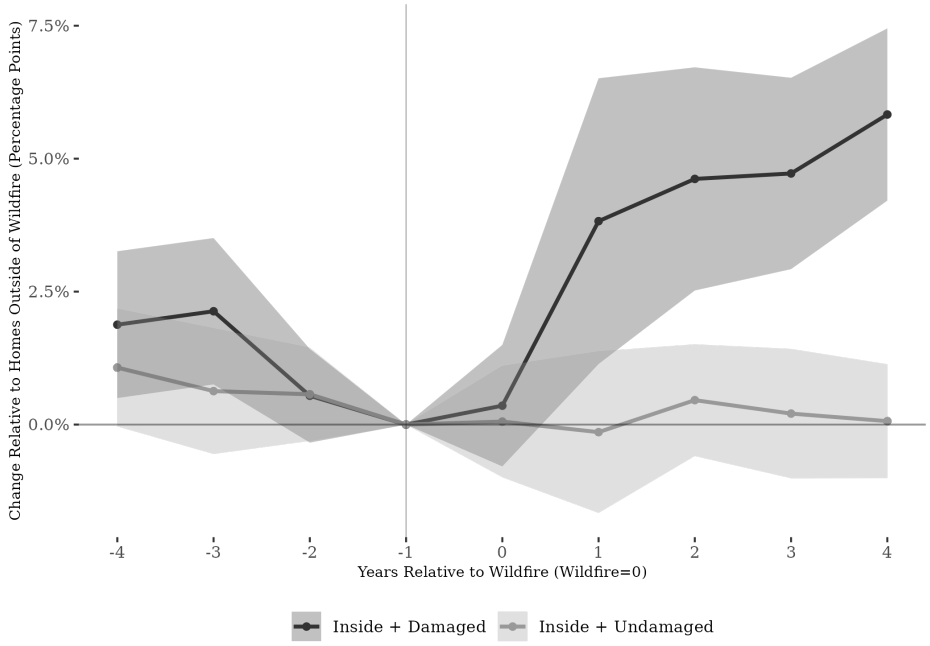
## 4.3 Rebuilding

We examine rebuilding activity for damaged properties using the likelihood of filing a building permit for new construction and of borrowing a construction loan. As shown in the top panel of Figure 6, rebuilding activity, measured by building permits, is low for at least two years after a property is damaged. Among damaged properties two years after a wildfire, only 10.8 percent have filed a new construction permit. While new construction permits may not pick up all reconstruction, 34.4 percent of damaged properties have filed any building permit of any type, providing an upper bound on rebuilding activity after a fire.

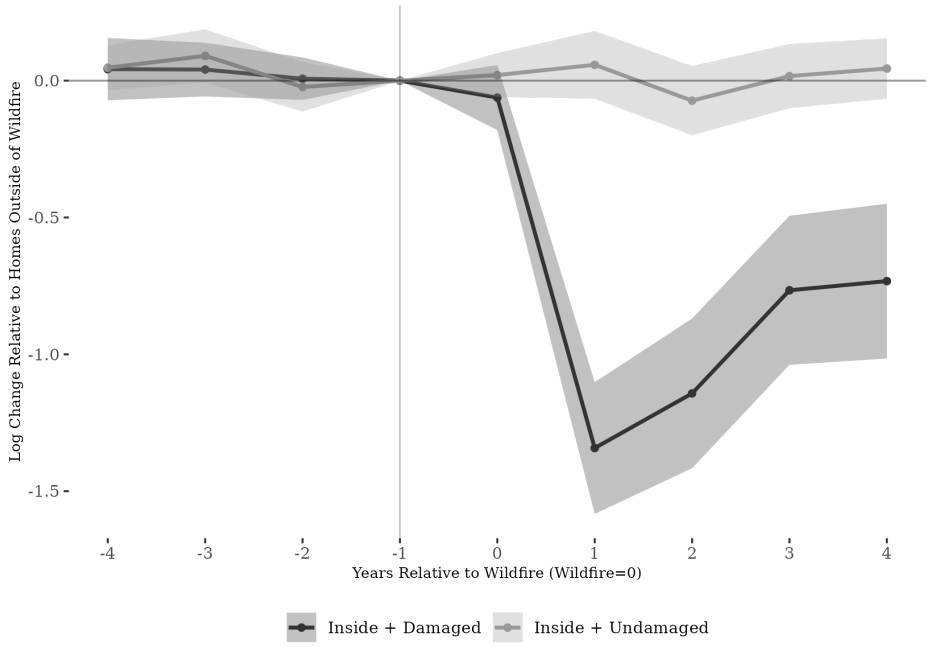
In the bottom panel of Figure 6, we estimate an annual event study with the likelihood of taking out a construction loan as the outcome and report the coefficients for damaged homes. The figure shows estimates from both the entire sample and the sample of homes that have not sold prior to the timing of the construction loan. There is a small overall uptick of construction loans for damaged properties, by up to 1.2 percent, in the four years after a fire. However, the prevalence of construction loans remains unchanged among homeowners who have held on to their property after incurring fire damage, increasing by less than 0.5 percent. This difference suggests that the new buyer of a damaged home borrows a construction loan to rebuild or restore the property.

While the damaged home's initial owner can rebuild without a construction loan due to the availability of insurance funds, the collection of results showing a lack of building

[25]We convert estimates of $\hat{\beta}_{1t}$, which measure log changes in price of -0.77 to -1.34, to percent changes.



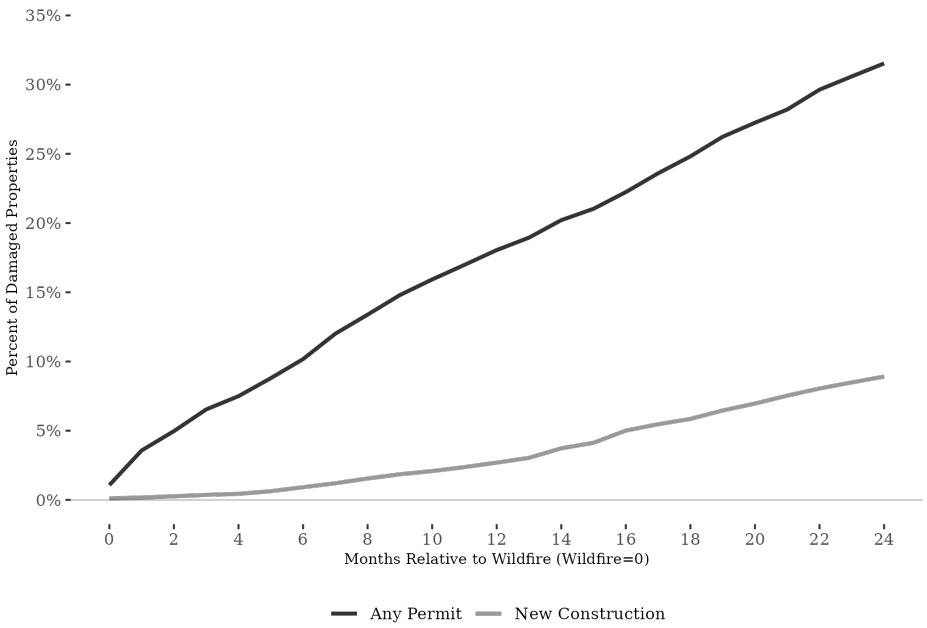

(a) Annual Probability of Sale



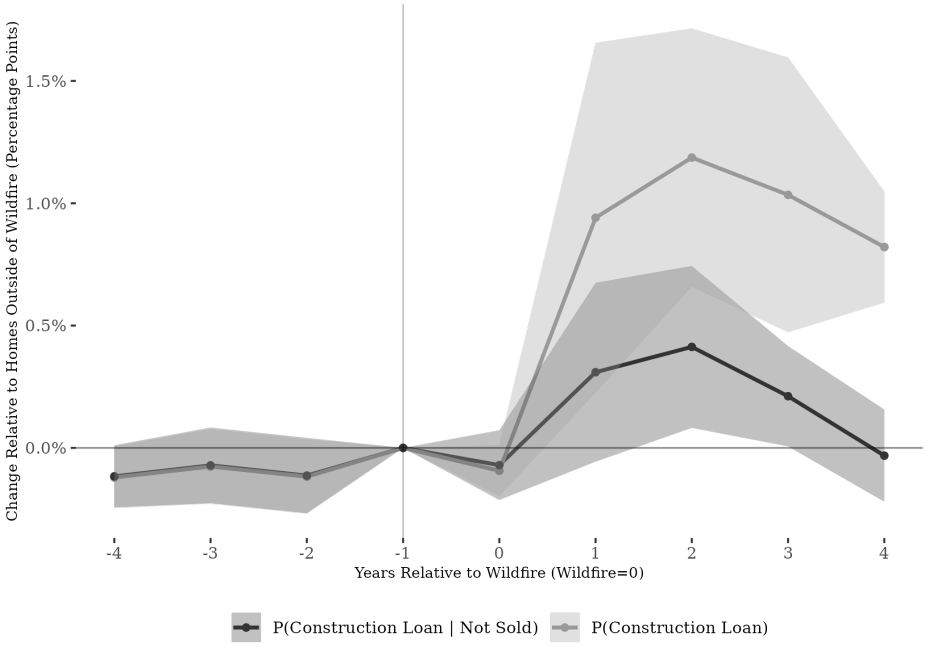

(b) Log Sales Price

Figure 5: Annual Sales Rates and Prices Around Wildfire Events

Note: Figure illustrates the coefficients for damaged ($\beta_{1t}$) and undamaged ($\beta_{2t}$) parcels estimated from an annual version of the event study specification in Equation 1. The outcomes for the top and bottom panels are an indicator for a sale and the log sale price, respectively. Regression controls for fire-year and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.




(a) Building Permits for Damaged Properties




(b) Construction Loans for Damaged Properties

Figure 6: Annual Sales Rates and Prices Around Wildfire Events

Note: The top panel illustrates the monthly trend in building permits among damaged properties after a fire. The bottom panel plots the coefficients from an annualized event study of damaged properties with the likelihood of borrowing a construction loan as the outcome. We estimate the regression for the entire sample of damaged homes and for damaged homes that have not sold. Regression controls for fire-year and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.

permits, significantly lower transaction price, and higher takeup rate of construction loans by the purchaser suggests that the original homeowner sells the damaged property before restoring it. Taken together, our findings support the following chain of events after a fire for damaged properties. Starting immediately after a fire, several damaged properties prepay their mortgage, possibly using their insurance claim. Few households file a new construction permit to rebuild. Over time, more damaged properties are sold at significantly lower prices, suggesting a land sale with an unrestored structure. Some new owners then take out a construction loan to rebuild the damaged property.

# 5 Insurance

So far, we find that the change in sales and refinances after fires is too small to explain the pronounced increase in prepayments and that households rebuild at low rates. In this section, we turn our attention to the role of property insurance in household's decisions after a fire. First, we discuss key features of the California insurance market that suggest the flow of insurance payments are consistent with the prepayment timeline. Second, using novel policy and claims data from the California Department of Insurance, we document that some households receive insurance settlements below coverage limits, which may limit their ability to rebuild.

## 5.1 Insurance Background and Claims Process

Wildfire damage is predominantly covered under commonly held homeowners insurance.[26] Typical policies include coverage for the replacement cost value (RCV) of the dwelling (denoted as Coverage A) and additional structures of the residence, personal property, living expenses during the loss of use of the residence, personal liability, and medical payments to

[26] California policy data show that there are 0.86 homeowner insurance policies per single family home, implying a much higher take up rate than flood insurance. One reason for high take up is that Fannie Mae and Freddie Mac require homeowners insurance for any mortgage that they purchase (see Fannie Mae's Seller's Guide Section B7-3-02 and Freddie Mac's Seller/Servicer Guide Section 4703.2).

others. Homeowners can also purchase extended replacement cost value (ERCV) coverage, which allows the settlement to exceed Coverage A to account for any increased costs of construction and compliance with local building codes after a fire. As a condition of having ERCV coverage, insurers require that the coverage limit A is equal to the RCV (Klein, 2023).

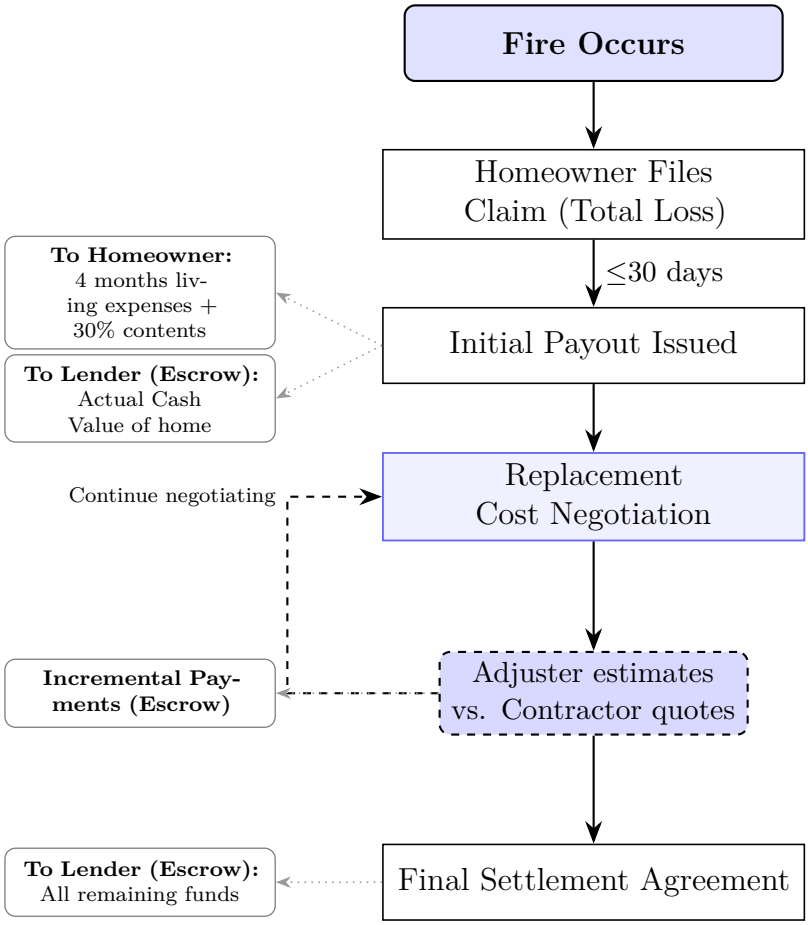



Figure 7: Insurance claims process timeline

Figure 7 summarizes the claims process, which often consists of a stream of incremental payments made by the insurer, not just one settlement check at the end of a negotiation. After a fire, the homeowner initiates the process by filing a claim with their insurer. For a total loss and a declared state of emergency, insurers advance four months of additional living expenses as well as 30% of a homeowner's contents limit without documentation.[27] Insurers

[27]This was codified effective January 1, 2021 by SB872. However, in previous disasters, including 2017 and 2018 wildfires, the California Insurance commissioner directed insurers to provide the same quick advances.

will also often advance the actual cash value (ACV) of the property, less depreciation, until the repair or replacement costs can be estimated and negotiated (United Policyholders, 2026; Klamm and Berline, 2025).[28] The property's ACV is typically not in dispute and insurers must pay the portion of claims not in dispute within 30 days (Merlin Law Group, 2024). The ACV check is deposited into an escrow account with the lender until the homeowner decides to repay their mortgage, rebuild their home, or move.

While the insurer's indemnity is not a function of the homeowner's ultimate decision, the final amount is a function of the negotiation between the homeowner and insurer. California residents with replacement cost coverage are entitled to the lesser of the full extended coverage limit and the quoted costs of rebuilding the home at the original site regardless of the household's decision to rebuild, buy, or newly build a home elsewhere.[29] Reaching an agreement on the actual replacement cost of the destroyed building can be an arduous process. Insurers often require extensive documentation and negotiation over the full scope of loss and associated building costs.[30] Furthermore, if a homeowner has decided not to rebuild, it can be especially difficult to get a contractor to spend time negotiating with an insurer over a construction project that will not happen.

Therefore, the homeowner will have the actual cash value of their home in an escrow account with their mortgage servicer/lender and will not receive additional funds until they reach an agreement on the indemnity amount with their insurer. The speed at which an agreement can be reached depends on a variety of factors including the insurer, labor shortages, and the homeowner. Some homeowners may quickly decide to buy or rebuild elsewhere, in which case they will have initial funds to pay off their mortgage while they negotiate for

[28] See Appendix A.1 for suggestive evidence for initial ACV payments and CA Ins Code §2051.5 (a) (2).

[29] See California Insurance Code §2051.5. If the homeowner decides to purchase a new house, the claim entitlement is also capped by the price of the house.

[30] After a claim, an insurance adjuster will estimate the insured losses. However, after large disasters, there are often adjuster shortages that can substantially delay the process. If homeowners want to dispute the insurer's estimate, they must get a contractor quote, which can also be delayed due to contractor shortages. When estimates differ, the insurer, homeowner, and contractor must negotiate over the differences and agree on the exact "scope of loss" and estimated costs.

additional funds with the insurer. Since homeowners are likely to have received the initial disbursement of the actual cash value of the home quickly, they may be able to prepay their mortgage quickly if they have decided not to rebuild.[31]

## 5.2 Policy and Claims Data

We gather data on anonymized fire policies and claims in California from 2018 to 2021 from the California Department of Insurance (CDI) to estimate homeowners' reimbursement gaps between coverage amounts that proxy rebuilding costs and actual received settlements.[32] The data include information on all policies in force in a given year for insurers that write $10 million or more in premiums for property or homeowners insurance, covering over 98 percent of the voluntary market. The data are repeated cross sections of policies that can be linked to the claims made on each policy. The policy data include coverage type, limits, premiums, deductibles, and start and end dates, and the claims data include the type, date, and amount of loss. Notably, the date reveals the timing of the loss but not the timing of the claim filing or payment. The data does not provide geographic details beyond the property's zip code and does not allow us to identify the insurer associated with each policy.

The claims data collected from insurers includes loss records as of January 31 of the reporting year. While we refer to this field as "claims", each loss record is the insurer-reported amount of claims paid plus case loss reserves. Case loss reserves are additional funds set aside by the insurer that reflect their estimated cost to cover the remainder of a partially settled claim. Concerns about variation in settlement timelines may affect the share of incurred losses allocated to actual paid claims versus case loss reserves, but should not substantially

[31] See Appendix A.2 for an analysis of the January 2025 Southern California wildfires which included the large Palisades and Eaton wildfires. We find that within two months, 80% of claims had been filed, 49% of claims were partially paid, and 29% of total payouts had been disbursed. The quick filing and initial payment align with the rapid prepayment spike we find in our mortgage data.

[32] The data are collected through Public Records Request GOVR-2023-00074. Every other year, the CDI collects policy and loss data from insurers for the previous two years through the Personal Property Experience (PPE) data call. Our data request includes two such data calls that cover information from 2018 to 2021. More information about the data collection and aggregate reports derived from the data are available at https://www.insurance.ca.gov/01-consumers/200-wrr/WildfireRiskInfoRpt.cfm.

affect the total amount of insurer-estimated incurred losses.[33] We focus on the sample of fire coverage policies for single units through homeowner or standalone dwelling-fire insurance.[34] For each claim, we assign the total dollars paid across all months to the month of the incurred loss to account for settlements received over multiple payments. Lastly, we only include fire-related and catastrophic loss claims on the primary structures (type A).[35]

Stylized facts from this data confirm that homeowners insurance is widely held and most insured households hold coverage at least as large as estimated replacement costs. First, we estimate there are 0.86 policies per single family home in zip codes with fire damage. Although we cannot directly estimate whether a property has insurance or not, this policy to property ratio suggests a high take-up rate for homeowners insurance that contrasts with low flood insurance take-up, even where coverage is mandated in Special Flood Hazard Areas (Wagner, 2022). Furthermore, we find that 88 percent of California homeowners insurance policies hold extended RCV coverage. Therefore, most insured households hold coverage at or above an insurer-provided estimate of the cost to replace their home. Such homeowners experiencing a total loss, such as home destruction from a wildfire, can expect to receive the lesser of actual rebuilding costs and expected rebuilding costs stated by their coverage limit.

## 5.3 Insurance Reimbursement Gaps

Using the policy and claims data, we construct a reimbursement gap that measures the shortfall between coverage limits and insurance claims. For each claim $i$ in zip code $j$ and month $t$, we calculate the reimbursement gap as the difference between claims and estimated

[33] Insurers' initial incurred loss estimates are relatively stable over time. Aggregate incurred losses in our 2018-2021 sample were only 4% less than the aggregate incurred losses reported with the most recently published data. Insurers have strong incentives to estimate loss reserves accurately because it affects their solvency and their financial ratings.

[34] This restriction matches our mortgage repayment analysis sample of single-family residences.

[35] In total, we exclude 10 percent of policies: guaranteed replacement cost coverage (GRCV) policies (less than 4 percent of policies), outliers in coverage limits (top and bottom 2.5 percent within zip code-year), and outliers in claim-to-limit ratios (top 2 percent). We further discuss these outliers and conduct robustness checks by adding back GRCV policies and outliers in the Appendix (Table A.3).

replacement cost value, as measured by the Coverage A limit:

$$\text{Reimbursement Gap}_{ijt} \equiv \text{Coverage A Limit}_{ijt} - \text{Claim}_{ijt}.$$

A notable limitation of these data is the lack of property address. While we observe each policy's zip code, we are not able to link the policy or claim to a specific property in the damage inspection data from CAL FIRE. Therefore, we do not observe the coverage or claims for a specific destroyed property. Instead, we identify the likely set of claims made on the destroyed properties in a zip code in any particular month.

First, we rank all claims from highest to lowest within zip code $j$ and month $t$. Then, we keep the top $N_{jt}$ claims, where $N_{jt}$ is the number of properties destroyed in zip code $j$ and month $t$. For example, if there were 100 properties destroyed in a zip code from a specific fire, we keep the 100 largest claims in that zip code in the month of the fire under the assumption that the largest fire claims result from properties destroyed by wildfires.

Restricting the sample of claims by the number of destroyed properties conservatively measures reimbursement gaps. If a fire claim due to reasons other than a wildfire exists in our sample, it replaces a claim resulting from a wildfire that is of a lower amount.[36] Therefore, any bias for insured homes will overestimate the total claims paid resulting from the wildfire. We also assume that the Coverage A limit is the actual replacement cost of a destroyed property, even though insurer-estimated RCV that dictates coverage limits is often found to be much lower than the actual costs of rebuilding (Dixon et al., 2018; Klein, 2018, 2023; Cookson et al., 2024). Our estimate of reimbursement gaps may be lower than the true deficit between the insurance settlement and the costs of rebuilding.

Our results suggest a high prevalence of possible reimbursement gaps. Between 2018 and 2020, approximately 33 percent of claims faced a possible reimbursement gap (Table 3). The average gap is \$64,316, or 15 percent, of coverage limits. That is, the largest claims

[36] Examples of fire claims beyond a specific wildfire include a kitchen fire occurring at the same time as a wildfire, fraudulent claims, or claims on undamaged properties requiring hazardous debris removal.

in zip codes where we observe fire damage received 85 cents per dollar of their respective Coverage A limits. For the claims that had any gap, the average gap jumps to 79 percent of coverage limits. The large difference between the average gap and conditional gap captures the bimodal distribution of claims shown in Figure A.12, where some households receive large settlements relative to their limits and other households receive low settlements. This finding suggests that comparing average claims to average coverage limits would understate important distributional consequences arising from the sizable share of households facing possibly large reimbursement gaps. The prevalence of reimbursement gaps and conditional gaps are similar to results from surveys of affected homeowners that show 42 percent to 66 percent of respondents would not be fully covered to replace or rebuild their home, with average deficits ranging from $163,000 to $375,000 (United Policyholders, 2022).

Table 3: Reimbursement Gaps

| | All | 2018 | 2019 | 2020 |
|---|---|---|---|---|
| Average Gap (share of limit) | 0.152 | 0.126 | 0.198 | 0.247 |
| Average Gap ($) | 64316 | 50635 | 116728 | 126410 |
| Any Gap (share) | 0.328 | 0.304 | 0.509 | 0.438 |
| Average Gap, if any gap (share of limit) | 0.789 | 0.809 | 0.640 | 0.734 |
| Average Gap, if any gap ($) | 456865 | 458171 | 400424 | 454585 |
| Number of Claims | 10248 | 8391 | 53 | 1804 |
| Number of Zip-Months | 96 | 28 | 11 | 57 |

Note: Table reports reimbursement gaps, which is calculated as the amount by which the settlement is below the Coverage A limit. Gaps that are represented as shares are dollar weighted by the coverage limit. We include the largest $N_{jt}$ claims in zip code $j$ and month $t$, where $N_{jt}$ is the number of observed wildfire-destroyed properties, to identify the set of claims associated with wildfire damage. Dollar amounts are deflated to 2020 USD.

One potential concern with this method is that we may assign a paid claim to a home that does not carry insurance. This concern is mitigated as 86% of properties in our sample's zip codes carry insurance. We adjust our reimbursement gap analysis by restricting the sample to the top $N_{ijt}^{Ins}$ claims in a zip code, where $N_{ijt}^{Ins}$ is the number of insured properties

destroyed in zip code $j$ and month $t$.[37] These results show that 25 percent of households face possible reimbursement gaps (Table A.2). While the average reimbursement gap shrinks, the average gap conditional on any gap remains the same. Results are also similar if we include guaranteed replacement cost policies, assuming their gap is zero, and outliers (Table A.3). While the average gap is sensitive to our assumptions, the distributional result that many affected households (25 to 33 percent) possibly face large reimbursement gaps is robust. Appendix A.1 shows further analysis of our insurance data.

There are several possible reasons why households may face reimbursement gaps. First, the frictions in the insurance process noted previously may incentivize households to agree to quicker and lower settlements after the initial ACV advance payment in order to reduce hassle costs. These hassle costs include obtaining multiple quotes for a reconstruction cost estimate, the length and uncertainty of the claim settlement process, and several stages of negotiations with the insurer.[38] Second, insurers may be paying the full coverage limit, but subtracting a range of deductions. For example, in the case of a new home purchase, the insurer may argue that they can deduct the value of the land from the purchase price since the insurance payout should only be limited to structures.[39] The "coinsurance" clause presents another deduction, which allows the insurer to pro-rate the amount that they pay out in proportion with the amount the homeowner was underinsured by.[40]

We find additional reasons for reimbursement gaps to be less plausible. First, the homeowner may have an actual cash value (ACV) policy and receive their full entitled payout

[37] We calculate the number of insured properties destroyed using the zip code insured share and number of properties destroyed.

[38] While outside our analysis period, Commissioner Lara issued a 2025 Bulletin that highlights cases where some homeowners received advice from insurance adjusters that their payout would depend on whether they rebuilt their house or not. The Bulletin states this advice is not aligned with California law, noting that "some insurance companies and/or adjusters appear to be misinforming policyholders". See https://www.insurance.ca.gov/0250-insurers/0300-insurers/0200-bulletins/bulletin-notices-commiss-opinion/upload/Bulletin-2025-2-Wildfire-Consumer-Protections-and-Advanced-Payments.pdf.

[39] As of Jan 1, 2021, this type of deduction was explicitly prohibited. Even before 2021, there was a strong case to be made for disallowing insurers from deducting land values (Lara 2019).

[40] See Section 1 - Conditions; C. Loss Settlement 2b of the 1999 ISO HO-3 form and Merlin Law Group (2021) for an example. We do not observe which companies may or may not include this clause in our data.

that are significantly lower than coverage limits. Our sample shows that 99% of the CA insurance policies are noted as RCV or better (i.e., RCV, extended RCV, or guaranteed replacement). This is expected as GSE rules require insurance policies to be settled on an RCV basis. Second, households may be overinsured – that is, the actual reconstruction costs observed in claims may be below the estimated replacement costs represented by policy limits. However, a 2008 Market Conduct Exam conducted by the CA Department of Insurance found that the recommended insurance coverage understated actual reconstruction costs over 80% of the time and even when homeowners purchased extended replacement cost coverage, 57% of the policies were still underinsured relative to the cost of rebuilding their homes.[41] In light of these reports and surveys, we find it unlikely that when a claim is paid less than the coverage limit, that it is because the homeowner had a coverage limit that was on average $450,000 more than the actual cost to rebuild their home.

# 6 Implications

Our estimates of possible reimbursement gaps have large implications for household equity, public finances, and mortgage market participants. Households may experience a sharp decrease in wealth if insurance claims do not cover rebuilding costs. Reimbursement gaps would reduce liquidity for households to rebuild, which reduces property tax revenues for localities due to low tax assessments. As shown in our previous results, households increase prepayment, which may shift post-disaster default risks to prepayment risks for lenders and investors in mortgage markets. This section quantifies the implications for these various agents.

[41] See Association of California Insurance Companies v Dave Jones (2017). While this Exam is from 2008, post-disaster surveys by United Policyholders have continued to find many homeowners do not have sufficient insurance funds for rebuilding.

## 6.1 Household Equity

For households, we calculate a back-of-the-envelope change in equity (focusing on the difference between their loss and their claim) from before to after the fire based on received insurance claims. We then compare this change in household equity to the resulting change in the counterfactual where households receive a full insurance payout. The difference in the equity change relative to the counterfactual quantifies any financial loss to the household due to possible reimbursement gaps.

A full insurance payment of the coverage limit is a natural counterfactual as California residents with replacement cost coverage are entitled to the lesser of the full coverage limit and the quoted costs to rebuild the damaged home in the original location.[42] The counterfactual change in equity from before to after the fire for households that receive the full payment is therefore represented in Equation 2. We assume that home value decreases by the amount of damage, $D_{ijt}$, for homeowners that do not rebuild and homeowners that do rebuild incur reconstruction costs, $D_{ijt}$, to restore the damaged value.

$$\Delta e^*_{ijt} = \min\{(1 + \text{ERCV}_{ijt}) \cdot \text{Coverage A Limit}_{ijt}, D_{ijt}\} - D_{ijt}. \tag{2}$$

However, households may not receive their entitled claim. Therefore, the realized changed in equity from before to after a fire after accounting for actual received claims is

$$\Delta e_{ijt} = \text{Claims}_{ijt} - D_{ijt}. \tag{3}$$

The counterfactual equity change in Equation 2 nests two possible outcomes for households: (1) the replacement cost is below the coverage limit, in which case the homeowner is made whole and suffers no loss or (2) the replacement cost is above the coverage limit and the homeowner has to cover the difference, which is their loss.

[42] See California Insurance Code §2051.5.

We do not observe the actual replacement cost of the property. Instead we assume that insurers should pay out at least the full limit of the insurance policy for the sample of destroyed homes in our analysis. We consider this assumption to be reasonable as most coverage limits are supposed to be for full replacement cost coverage. Additional evidence shows that these replacement cost estimates, and therefore coverage limits, may be biased downward (Klein, 2018, 2023). Under these assumptions, we calculate a household's equity shortfall as the amount of equity lost when comparing received claims to the counterfactual scenario where they receive their entitled payout of the full coverage limit after a total loss.[43]

$$\text{Equity Shortfall}_{ijt} \equiv \min\{\Delta e_{ijt} - \Delta e^*_{ijt}, 0\} = \min\{\text{Claims}_{ijt} - \text{Coverage A Limit}_{ijt}, 0\} \quad (4)$$

This calculation assumes that, for claims without a reimbursement gap, households receive their entitled payout and thus face a zero shortfall. This calculation is likely underestimating the shortfall for households receiving claims exceeding limits, as we assume their settlement equals the full replacement cost. For claims with reimbursement gaps, the shortfall between the actual and counterfactual equity change is simply the reimbursement gap.[44] As shown in Table 4, household gross equity is \$151,650 to \$233,457 lower than if they received the full coverage limit, which measures insurers' ex-ante estimates of rebuilding costs.[45]

## 6.2 Local Property Tax

In addition to reducing household wealth, possible reimbursement gaps have broader public finance implications because they may delay rebuilding if homeowners do not have enough

[43] Extended replacement cost coverage (ERCV) may exceed rebuilding costs. We provide two separate estimates of this shortfall comparing received claims to the Coverage A limit and the ERCV limit.

[44] Insurance premiums and deductibles cancel out in Equation 4 as the shortfall is calculated conditional on the same contract. The contract features are constant from before to after a fire and across the realization (actual claims received) and the counterfactual (entitled payout of coverage limit).

[45] Table A.5 shows that the shortfalls are identical even if we include the sample of GRCV policies and assume they have no shortfall. Even adjusting for the number of insured properties in each zip code leads to large gross equity losses of \$107,840 to \$176,755 (Table A.6).

Table 4: Impacts on Household Equity

| | | Shortfall Relative to: | |
|---|---|---|---|
| | $N$ | Coverage A ($) | ERCV ($) |
| All | 9,090 | -151,650 | -233,457 |
| 2018 | 8,178 | -138,058 | -215,702 |
| 2019 | 49 | -197,951 | -271,451 |
| 2020 | 863 | -277,821 | -399,551 |
| Camp Fire Only | 6,886 | -71,773 | -122,377 |
| Excluding Camp Fire | 2,204 | -401,210 | -580,507 |

Note: Shortfall amounts are calculated from Equation 4. We assume the counterfactual "full" payout is capped by either the Coverage A limit or the ERCV. Each row provides the same calculation for different subsamples of our claims data. We include the largest $N_{jt}$ claims in zip code $j$ and month $t$, where $N_{jt}$ is the number of observed wildfire-destroyed properties. Dollar amounts are deflated to 2020 USD.

money. Property damage reduces the tax base, which remains low until rebuilding occurs. We quantify this tax impact by examining public records data on residential property tax revenues for destroyed homes.

First, Figure 8 shows residential property tax collections for destroyed homes. All amounts are deflated to 2020 US Dollars. Relative to the year of the fire, tax revenues for destroyed homes are $50.6 million to $55.3 million lower per year for the three years following the fire, representing a 51.2 to 56.9 percent reduction in annual tax collections for destroyed homes.[46] The reduction in tax revenues is fully explained by a decrease in assessed values as tax rates remain unchanged from before the fire to after the fire.[47] Furthermore, the drop in tax revenues collected persists for at least three years after the fire. The large and prolonged drop in tax revenue for destroyed properties is consistent with the previously discussed lack of rebuilding.

Second, we show that possible insurance reimbursement gaps are associated with the drop in tax revenue among destroyed homes. We estimate a zip code level regression (Equation

[46] Liao and Kousky (2022) find that tax revenues increased after wildfires. Our results show lower revenues likely because our sample only includes destroyed homes while their sample includes municipalities overlapping fire perimeters. Tax revenue increases for undamaged homes may offset tax losses from destroyed homes.

[47] The average annual tax rate for destroyed properties is between 1.13 to 1.17 percent of assessed value from three years before to three years after the fire.

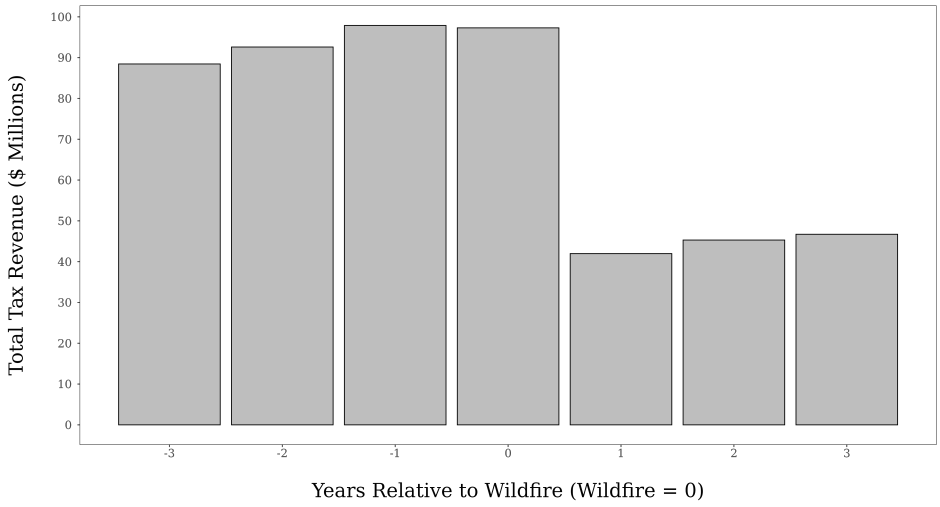



Figure 8: Property Tax Revenue for Homes Destroyed by Wildfires

Note: Figure illustrates the total residential property tax collected for our sample of damaged properties.

5) to test whether tax revenue from damaged homes varies after a fire by the percentage of claims in zip code $j$ facing a possible reimbursement gap between the claim and the coverage limit. We include zip code fixed effects and a zip code specific time trend to account for factors such as local house price appreciation. The time variable is measured in years relative to the fire and $\beta_s$ can be interpreted as the marginal increase in residential property tax revenue per destroyed property associated with a marginal increase in the rate of claims with possible reimbursement gaps.

$$\ln(\text{Property Tax}_{jt}) = \sum_{s=-5;s\neq-1}^{s=4} \beta_s \text{Reimbursement Gap}_j \times \delta_{jt}^s + \lambda_j + \lambda_t + \epsilon_{jt} \tag{5}$$

Figure 9 shows that zip codes with higher rates of possible reimbursement gaps face a larger decrease in tax revenues per property for the first two years after a fire. A one percentage point higher share of claims facing a possible reimbursement gap is associated with a 0.65 to 0.82 percent decrease in tax revenues per year. These elasticities imply that the average gap rate of 33 percent is associated with a 23.6 percent reduction in tax revenues for

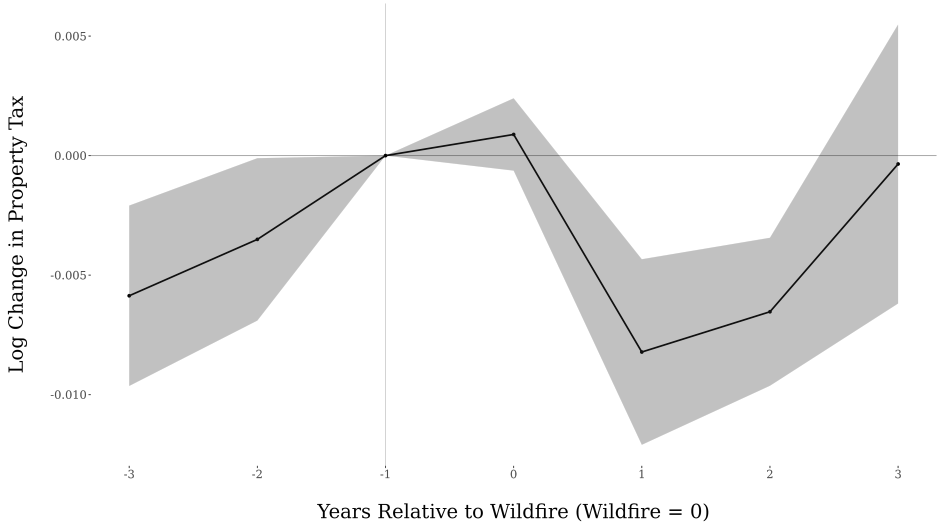



Figure 9: Impact of Insurance Reimbursement Gaps on Tax Revenues

Note: Figure plots the coefficient $\beta_s$ from Equation 5, with log tax revenue per property as the outcome variable. Regression controls for zip code fixed effects and zip-specific time trends. Shaded regions indicate 95% confidence intervals clustered by zip code.

damaged homes in the year after the fire relative to the counterfactual of no reimbursement gaps.

## 6.3 Lenders, Servicers, and Investors

For lenders, servicers, and investors in mortgage-backed securities, natural disasters present both default and prepayment risks. Even if damaged properties rarely default after a wildfire in our loan sample, the increase in prepayment can lead to losses. In the case of a prepayment, lenders lose anticipated interest revenue and servicers lose recurring servicing fees. Moreover, the temporary reduction in housing supply due to fire damage would impact the lender's ability to replace the mortgage in its portfolio. While investors in mortgage-backed securities are protected by the geographic diversification of their portfolios, prepayments reduce the cash flow duration and present the most salient risk for these securities (Fuster et al., 2025).

However, total losses to lenders may be modest given the small number of mortgages affected by wildfires relative to the lender's portfolio. While we cannot calculate the total losses lenders, servicers, and investors face, we test whether the marginal damaged home in a

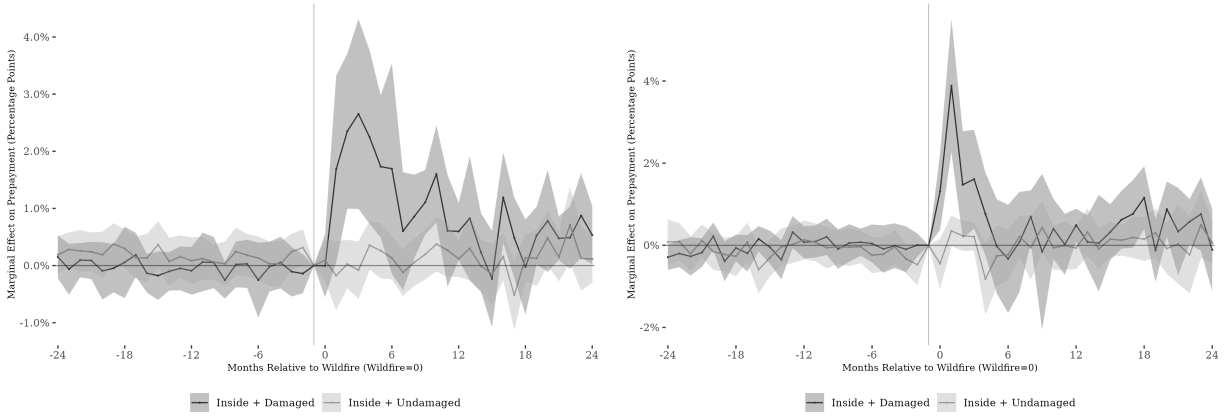
mortgage portfolio would increase the loss of interest revenue. We estimate heterogeneous effects on prepayment using the specification in Equation 6 with the loans' origination interest rates as $Z_{ift}$. Banks would prefer borrowers with lower interest rates prepay to minimize lost interest revenue and to potentially add a new mortgage with a marginally higher interest rate. The estimate of $\hat{\gamma}_{1t}$ in panel (a) of Figure 10 opposes bank preferences, as loans with higher interest rates have higher likelihoods of prepayment after a fire. Loans with a 1 standard deviation higher interest rate have a 2.7 percentage point higher likelihood of prepayment.

$$
\begin{aligned}
Y_{ift} = & \sum_{t=-24;t\neq-1}^{t=24} \beta_{1t} Damaged_{ift} + \sum_{t=-24;t\neq-1}^{t=24} \beta_{2t} InsideFireZone_{ift} \\
& + \sum_{t=-24;t\neq-1}^{t=24} \gamma_{1t} Damaged_{ift} \times Z_{ift} + \sum_{t=-24;t\neq-1}^{t=24} \gamma_{2t} InsideFireZone_{ift} \times Z_{ift} \\
& + \lambda_{ft} + \lambda_{ft} \times Z_{ift} + \lambda_i + \epsilon_{ift}.
\end{aligned} \tag{6}
$$

This pattern could still be beneficial to lenders, servicers, and investors if higher interest rates are correlated with borrower risk, which would result in a lower risk borrower pool after high-interest mortgages prepay. Panel (b) of Figure 10 does not support this hypothesis and shows that less risky borrowers with higher credit scores are slightly more likely to prepay, resulting in a higher risk borrower pool.[48]

These results suggest that prepayment after wildfires is not beneficial for lenders, servicers, and investors, as they face losses in interest revenue. The large difference between the mortgage performance effects of property damage and the effects of disaster spillovers also has implications for lenders' risk modeling. As we find small spillover effects and large effects of direct property damage, lenders may manage risk by improving modeling and verification of property damages rather than exiting larger geographies that are affected by spillovers.

[48]We estimate the same heterogeneous effects for the delinquency outcome, as shown in Figure A.13. Borrowers with higher interest rates and lower credit scores are more likely to be past due by 90 days or more.




(a) Interest Rate × Fire Damage

(b) Credit Score × Fire Damage

Figure 10: Differences in Prepayment by Interest Rate and Credit Score

Note: Figure illustrates the coefficients for $Z_{ift}$ interacted with damaged ($\gamma_{1t}$) and undamaged ($\gamma_{2t}$) parcels estimated from the event study specification in Equation 6. Both plots present the differential impact on prepayment after the fire for loans with different interest rates (left panel) and borrowers with different credit scores (right panel). Coefficients estimate the marginal effect of a 1 standard deviation change in $Z_{ift}$. Regression controls for fire-month and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.

Our results suggest that targeted interventions to damaged properties, such as forbearance or support with the insurance claims process, may improve mortgage performance.

# 7 Conclusion

We construct a novel database that combines property-level damage inspections from 81 California wildfires with mortgage performance data to evaluate the impact of wildfires on mortgage repayment. This paper finds that 90-day delinquency rates and prepayment rates increase significantly for damaged properties immediately after a fire. In contrast, repayment for undamaged properties, including those inside the fire's perimeter, remains unchanged. The immediate 16 percentage point increase in prepayment cannot be explained by increased home sales or refinances after a fire. Using detailed home insurance policy and claims data, we find possibly large reimbursement gaps – properties destroyed by fires on average receive settlements that are lower than coverage limits, which proxy for insurer-estimated replacement costs.

Our results imply that wildfires present a larger risk to mortgage markets than previously estimated, result in lower rebuilding activity, and lead to financial losses for households and local governments. By accurately measuring property-level fire damage, we show that the direct effects of fire damage on repayment dwarf any spillover effects on undamaged homes inside the fire perimeter. Since our reduced-form parameter includes any protective effects of insurance and government aid, these risks for mortgages on damaged homes are potentially larger if safeguards weaken. Possibly large reimbursement gaps also imply that households are less financially resilient to natural disasters as they bear large negative wealth shocks from property damage. Extremely damaging physical risks combined with weaker-than-expected insurance protections have implications for several stakeholders.

For lenders, servicers, and investors, natural disasters pose default and prepayment risks. Even if damaged properties rarely default after a wildfire in our loan sample, the increase in prepayment can lead to lost interest revenue, higher damage verification costs, and changes to the borrower risk pool. For researchers, this study emphasizes the need to incorporate precise disaster damage measurement and strength of insurance protections in analyses of household finances after natural disasters. As 55 percent of properties and 58 percent of mortgages within the fire perimeter in our sample remain undamaged, using large geographic areas, such as fire perimeters, to proxy for property damage introduces substantial measurement error. Such measurement error and assumptions of full insurance protections would attenuate resulting estimates and understate the true impact of fires.

For policymakers, our findings indicate that damage mitigation efforts before fires may reduce the risks wildfires present to the mortgage market. Stronger wildfire building codes largely reduce the likelihood of property destruction (Baylis and Boomhower, 2026). After a fire, we observe persistently low rebuilding activity that leads to a reduced tax base and lower property tax revenues. We also find that areas with the highest rates of possible reimbursement gaps face the largest decreases in property tax revenues from damaged homes. Our results suggest that strengthening insurance protections and simplifying the claims

process after natural disasters would mitigate losses for homeowners, lenders, investors, and governments.

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# A Online Appendix

All results presented in the figures and tables of this section are derived from calculations based on the data described in Section 2 of the paper unless otherwise noted.

## A.1 Additional Insurance Analysis

### Outliers

**Zip-code aggregate** We aggregate claims to the zip-code level and plot the correlation between average claim amounts and average Coverage A limits in Figure A.14. Recall that California residents with replacement cost coverage who intend to rebuild or purchase a new home are entitled to receive the lesser of the full coverage limit and quoted replacement costs. Receiving claims at the full entitlement level which would suggest claims at or close to \$1 per dollar of coverage (the 45-degree solid black line). The figure shows that almost all zip codes, on average, face reimbursement gaps. Only five zip codes fall at or above \$1 of claims paid per dollar of coverage. Accounting for extended RCV, all (but one) zip codes' average payments are below the average coverage limit plus 20 percent additional coverage, as represented by the dotted line.

**Initial ACV payment** We consider the following hypothesis that could lead to large observed reimbursement gaps. In an effort to reduce costs, insurers commonly send initially low settlement offers that cover only the ACV, deducting the depreciation of the house's materials from replacement costs. To calculate the ACV, insurers require several details about the structure, such as the quality of finishes and size of cabinets. When such details are not available, insurance adjusters rely on aggregated characteristics of the zip code housing the property. Households may accept these offers because of a lack of information or unwillingness to enter the required negotiation process to secure a higher settlement based on the property's individual damages. In the event of large fires that require quick processing of claims, it is

likely that the dollar amount of claims would not vary greatly within zip codes. To examine this hypothesis, we estimate the following fixed-effects regression.

$$\text{Claim}_{ijt} = \beta \text{Coverage A Limit}_{ijt} + \gamma X_{ijt} + \lambda_j + \lambda_t + \epsilon_{ijt} \tag{7}$$

In Equation 7, $X_{ijt}$ includes additional extended replacement cost coverage and the policy premium rate per \$100,000 of coverage to account for policy quality. The fixed effects isolate the identifying variation to be as close to within-fire as our data allow by using zip code and month fixed effects. The $\beta$ from this regression can be interpreted as the additional dollar of claims a household with an additional dollar of coverage receives for the same fire and same quality of policy. If claims reflect the rebuilding costs of the individual properties, as estimated by the Coverage A limits, we expect $\hat{\beta}$ to be close to 1. Moreover, the fixed-effects estimate $\hat{\beta}$ should not differ substantially from the ordinary least squares (OLS) estimate unless the property's zip code is a critical component of determining the share of the coverage limit households receive in claims.

However, both of these hypotheses are rejected. Table A.4 shows that the fixed-effects estimate of the correlation between coverage limits and claims is weak, as $\hat{\beta}$ is statistically indifferent from zero. In addition, the correlation drops dramatically when introducing aggregated controls, such as extended RCV and policy price, and zip code fixed effects. These results support the hypothesis that the initial ACV settlement offer varies more by zip code characteristics than by the individual property replacement costs, as estimated by coverage limits.

## A.2 January 2025 Southern California wildfires

From January 7 to January 31, 2025, a number of wildfires affected the Los Angeles and San Diego areas of California. The most destructive of these wildfires were the Palisades and Eaton fires. On January 30, 2025, the California Insurance Commissioner published the

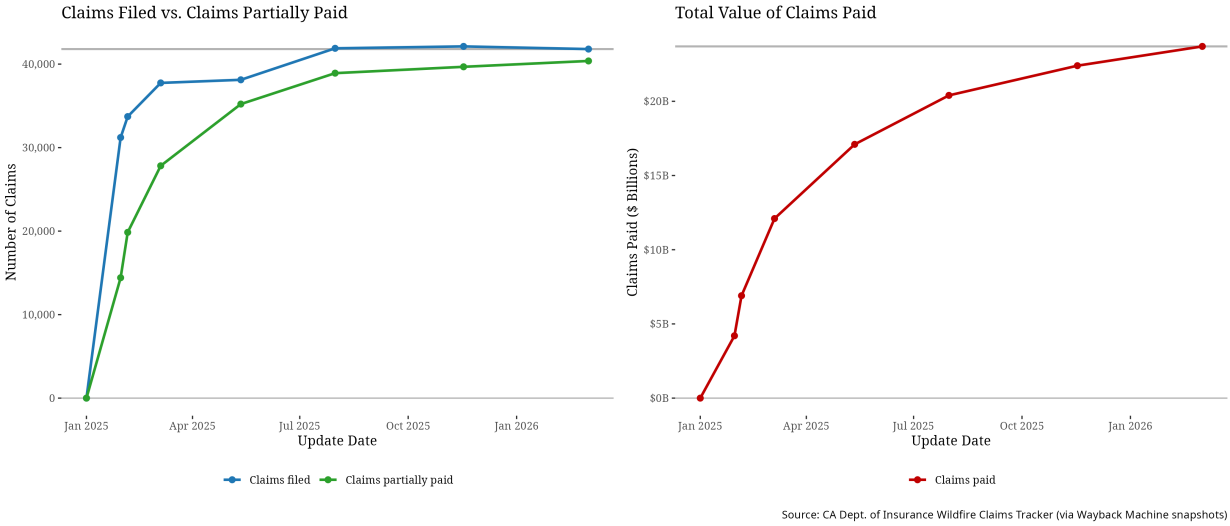
first-ever claims tracking system for the public to monitor insurance claims and payouts.[49] While the webpage only has the most recent snapshot, we collected previous snapshots from archived data to generate a time series of the claims and payout timeline for the January 2025 wildfires.[50]

While these wildfires are outside our analysis period, the tracker provides evidence that a substantial portion of claims are filed and at least partially paid quite quickly. In the month of the fire, 75% of claims were filed, 36% of claims were partially paid, and 18% of total payouts had been disbursed. Within two months, 80% of claims had been filed, 49% of claims were partially paid, and 29% of total payouts had been disbursed. These shares are computed against the total outstanding claims and claims payments as of March 2026. These statistics will likely change and total claims paid will continue to increase because homeowners have up to 36 months after a declared wildfire emergency to claim all of their funds (with six-month extensions granted for factors and delays outside of their control). However, even based on 15 months of post-wildfire data, there is substantial evidence that insurance claims are filed quickly and initial payouts are made quickly, which provides supporting evidence that the spike in prepayments we observe is driven by insurance payouts.

[49] See https://www.insurance.ca.gov/0400-news/0100-press-releases/2025/release011-2025.cfm for press release.

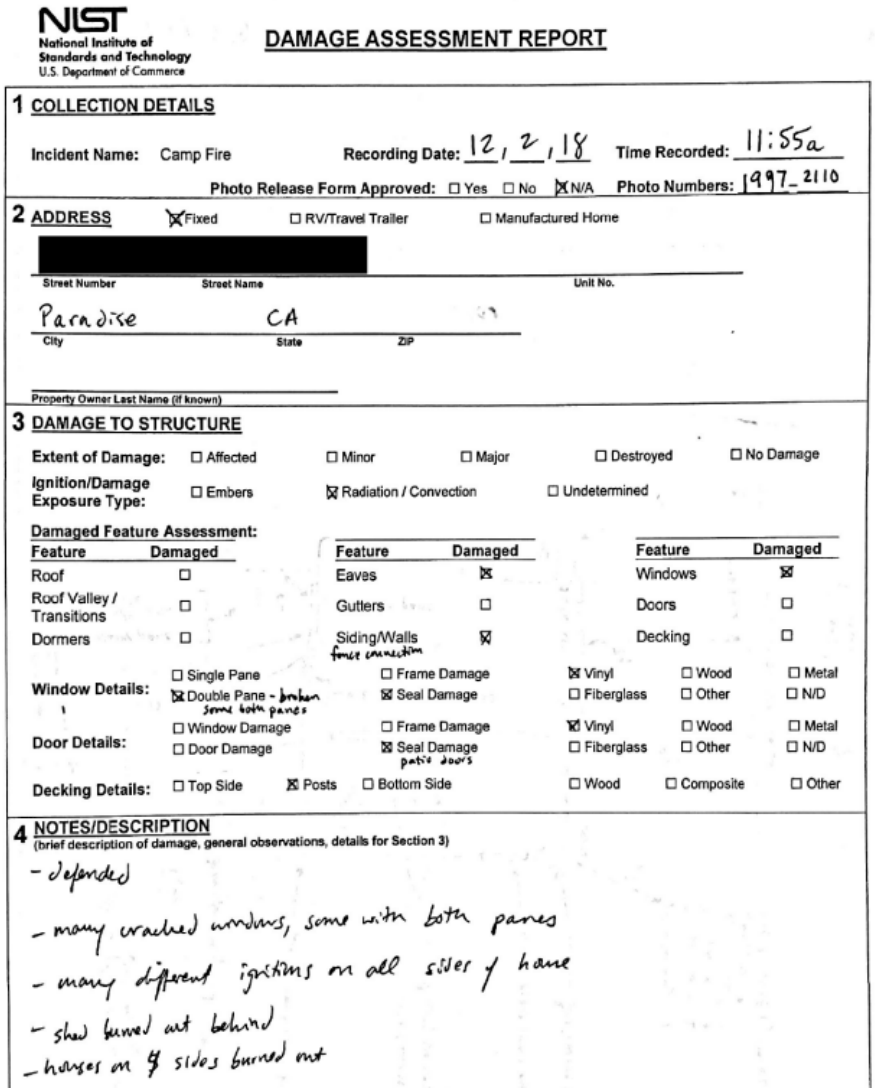
[50] Official claims tracker website is available at https://www.insurance.ca.gov/01-consumers/180-climate-change/Wildfire-Claims-Tracker.cfm. Historic snapshots were obtained from the Internet Archive's Wayback Machine.

Figure A.1: January 2025 California Wildfires Claims and Payments Timeline




## A.3 Additional Figures

Figure A.2: Example Damage Report for Camp Fire



NIST
National Institute of Standards and Technology
U.S. Department of Commerce

**DAMAGE ASSESSMENT REPORT**

**1 COLLECTION DETAILS**

Incident Name: Camp Fire  Recording Date: 12, 2, 18  Time Recorded: 11:55a

Photo Release Form Approved: ☐ Yes ☐ No ☒ N/A  Photo Numbers: 1997_2110

**2 ADDRESS**  ☒ Fixed  ☐ RV/Travel Trailer  ☐ Manufactured Home

[redacted]
Street Number  Street Name  Unit No.

Paradise  CA
City  State  ZIP

Property Owner Last Name (if known)

**3 DAMAGE TO STRUCTURE**

Extent of Damage: ☐ Affected ☐ Minor ☐ Major ☐ Destroyed ☐ No Damage

Ignition/Damage Exposure Type: ☐ Embers ☒ Radiation / Convection ☐ Undetermined

Damaged Feature Assessment:

| Feature | Damaged | Feature | Damaged | Feature | Damaged |
|---|---|---|---|---|---|
| Roof | ☐ | Eaves | ☒ | Windows | ☒ |
| Roof Valley / Transitions | ☐ | Gutters | ☐ | Doors | ☐ |
| Dormers | ☐ | Siding/Walls fence connection | ☒ | Decking | ☐ |

| | | | | | | |
|---|---|---|---|---|---|---|
| Window Details: | ☐ Single Pane | ☐ Frame Damage | ☒ Vinyl | ☐ Wood | ☐ Metal |
| | ☒ Double Pane - broken some both panes | ☒ Seal Damage | ☐ Fiberglass | ☐ Other | ☐ N/D |
| Door Details: | ☐ Window Damage | ☐ Frame Damage | ☒ Vinyl | ☐ Wood | ☐ Metal |
| | ☐ Door Damage | ☒ Seal Damage patio doors | ☐ Fiberglass | ☐ Other | ☐ N/D |

Decking Details: ☐ Top Side ☒ Posts ☐ Bottom Side ☐ Wood ☐ Composite ☐ Other

**4 NOTES/DESCRIPTION**
(brief description of damage, general observations, details for Section 3)

- defended
- many cracked windows, some with both panes
- many different ignitions on all sides of home
- shed burned out behind
- horses on 4 sides burned out

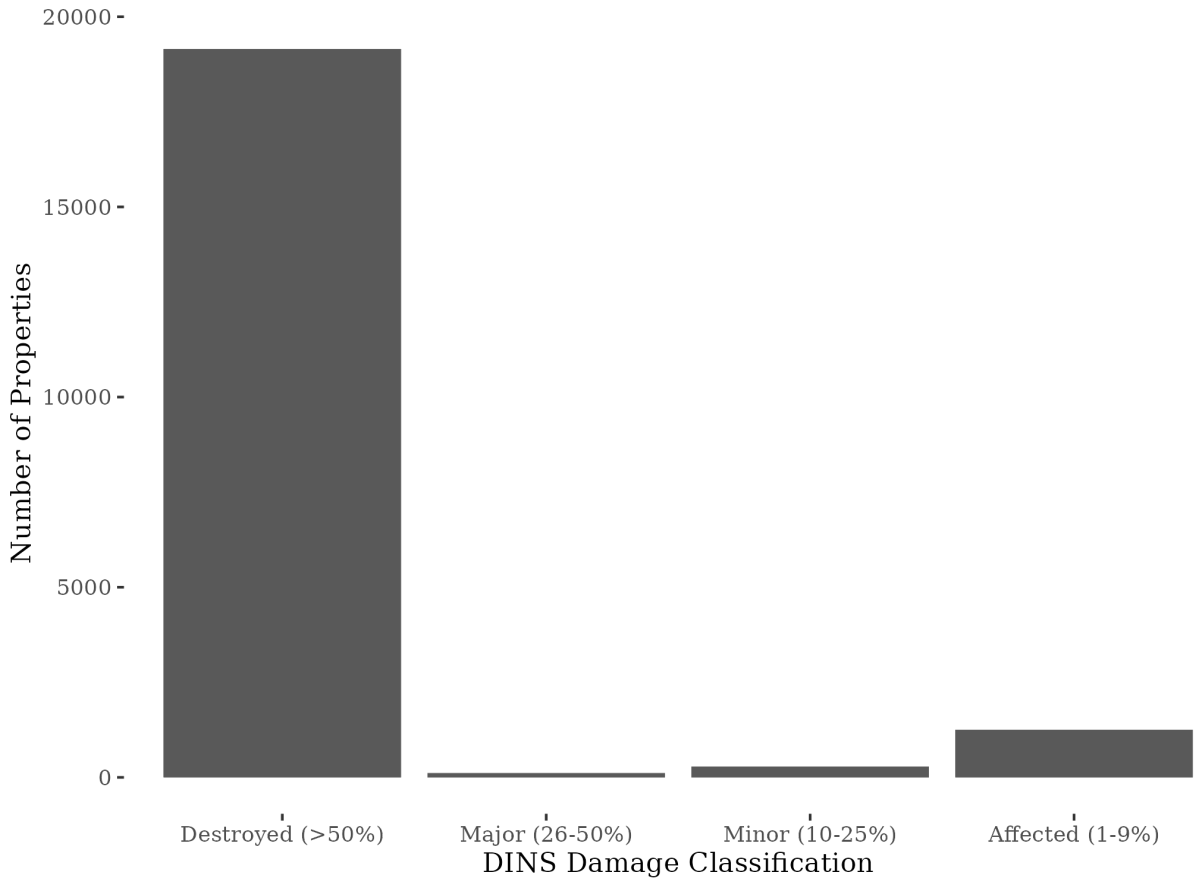
Source: NIST Investigation of the California Camp Fire, Fire Research Division, National Institute of Standards and Technology.

Figure A.3: Classification of Damage Categories

| Category of Damage | Definition | Examples |
|---|---|---|
| Affected (1-9%) | Minimal damage to the exterior and/or contents of the building. Building is habitable/usable and requires mostly cosmetic repairs. | Partially damaged shingles or siding, but roof structure is intact. Cosmetic damages such as paint discoloration, blistering or melted siding. Broken windows. Gutter damage. Damage to an attached structure like a deck, porch, carport, or patio cover. |
| Minor (10-25%) | Encompasses a wide range of damage that does not affect the structural integrity of the building. Building is not habitable/usable. | Nonstructural damage to roof components (e.g. roof covering, fascia board, soffit, flashing, and skylight). Nonstructural damage to the interior wall components to (e.g. drywall and insulation). Nonstructural damage to exterior components (e.g. door and windows. Substantial damage to exterior covering (e.g. siding, vinyl or stucco). Damage to mechanical components (e.g. furnace, boiler, water heater, HVAC, etc.). |
| Major (26-50%) | A building that has sustained significant structural damage and requires extensive repairs. Building is not habitable/usable. | Failure or partial failure of structural elements to include rafters, ceiling joists, ridge boards, etc. Failure or partial failure to structural elements of the walls to include framing, sheathing, etc. |
| Destroyed (>50%) | The building is a total loss, or damaged to such an extent that repair is not feasible. | Complete failure to major components (foundation, walls, roof, etc.). Two or more walls destroyed and roof substantially damaged. Only the foundation remains. The building will have to be torn down and rebuilt as it is unsafe. |

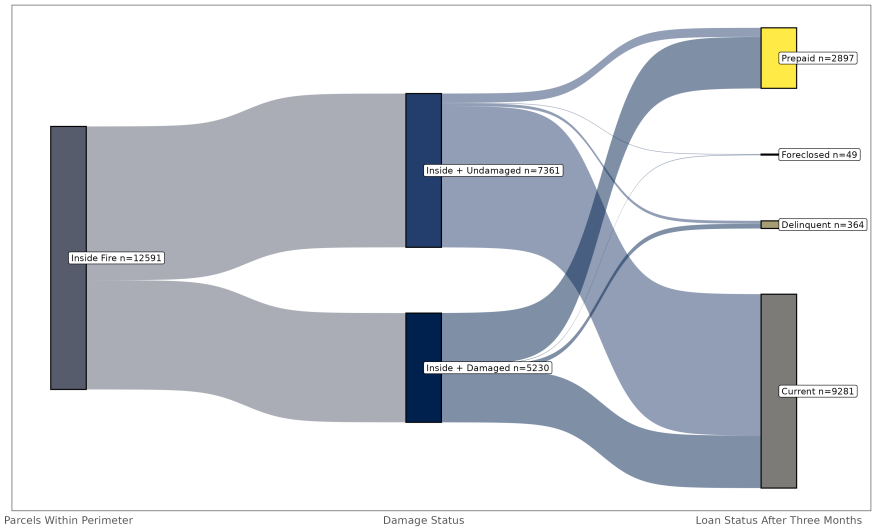
Source: Camp Incident Damage Inspection Report CABTU 016737

Figure A.4: Distribution of Damage Severity

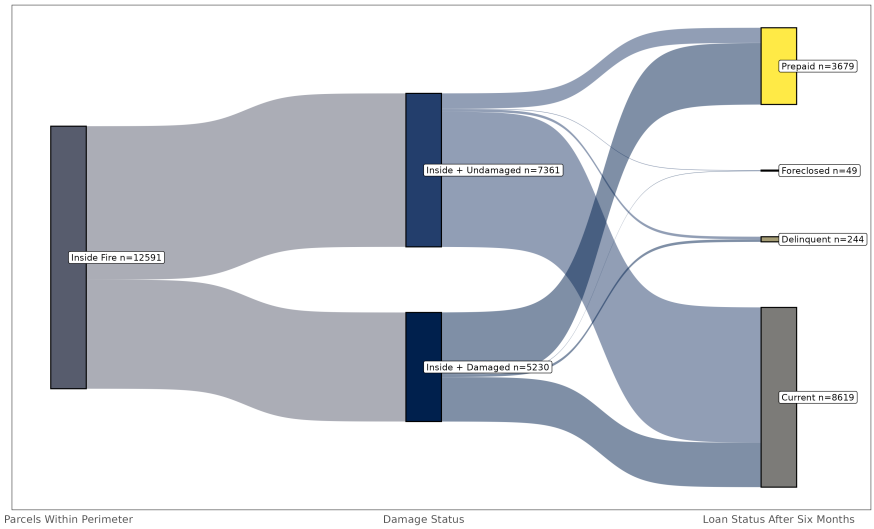



Note: Figure plots the distribution of damage categories reported in the DINS database.

Figure A.5: Post-Wildfire Loan Status Flows: Three and Six Months After Fire



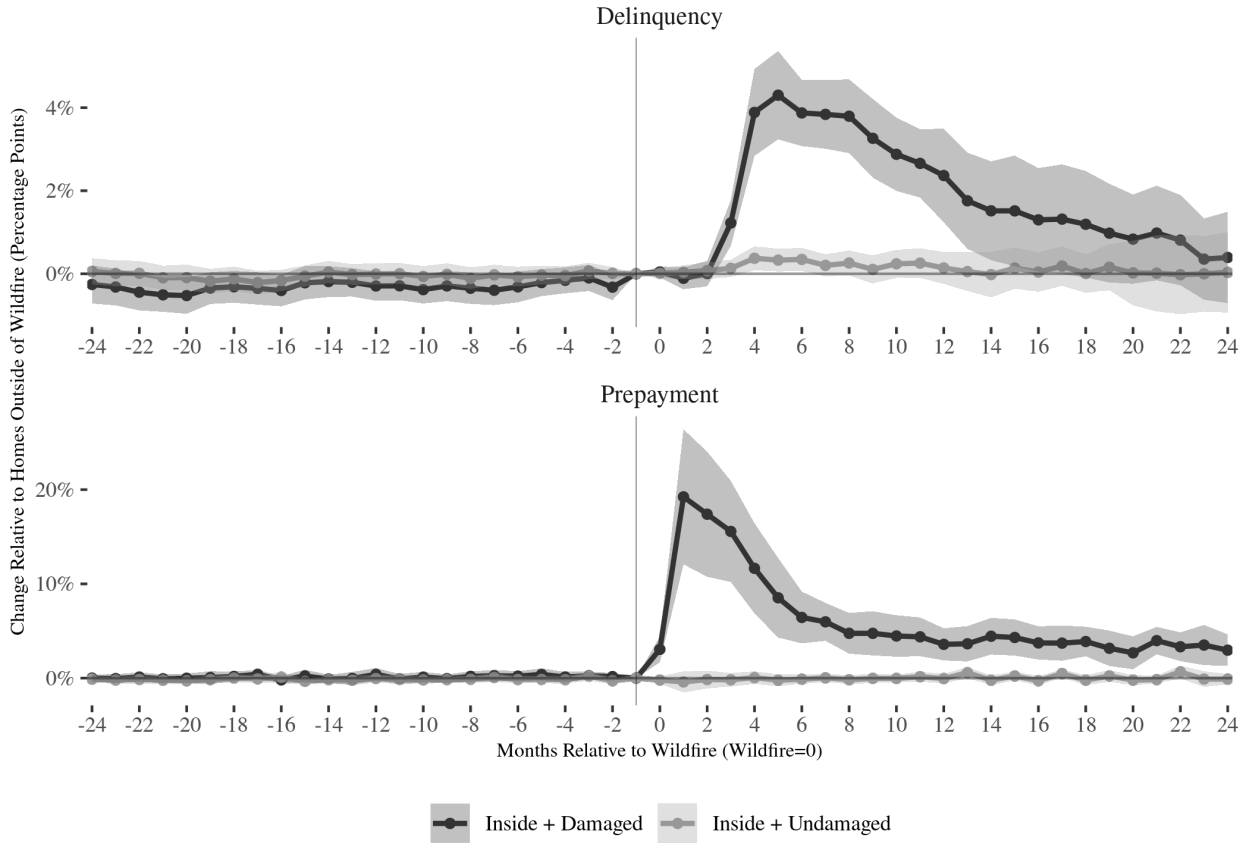

(a) Loan Status Three Months After Fire




(b) Loan Status Six Months After Fire

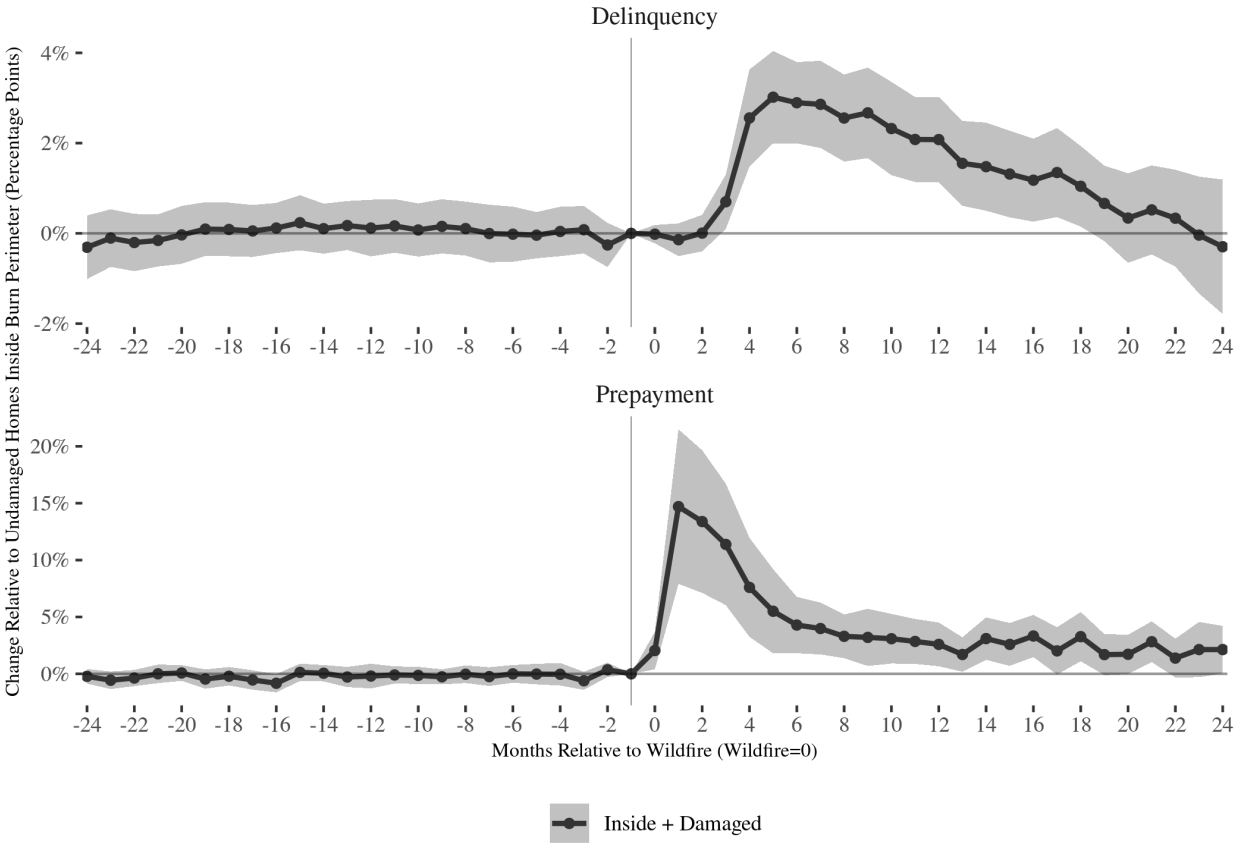
Note: Figure illustrates the decomposition of loans within wildfire boundaries based on whether or not the home was damaged by the wildfire. The top panel shows mortgage status three months post-fire (current, delinquent, foreclosed, or prepaid) while the bottom panel shows mortgage status six months post-fire.

Figure A.6: Robustness to Excluding Partial Damage



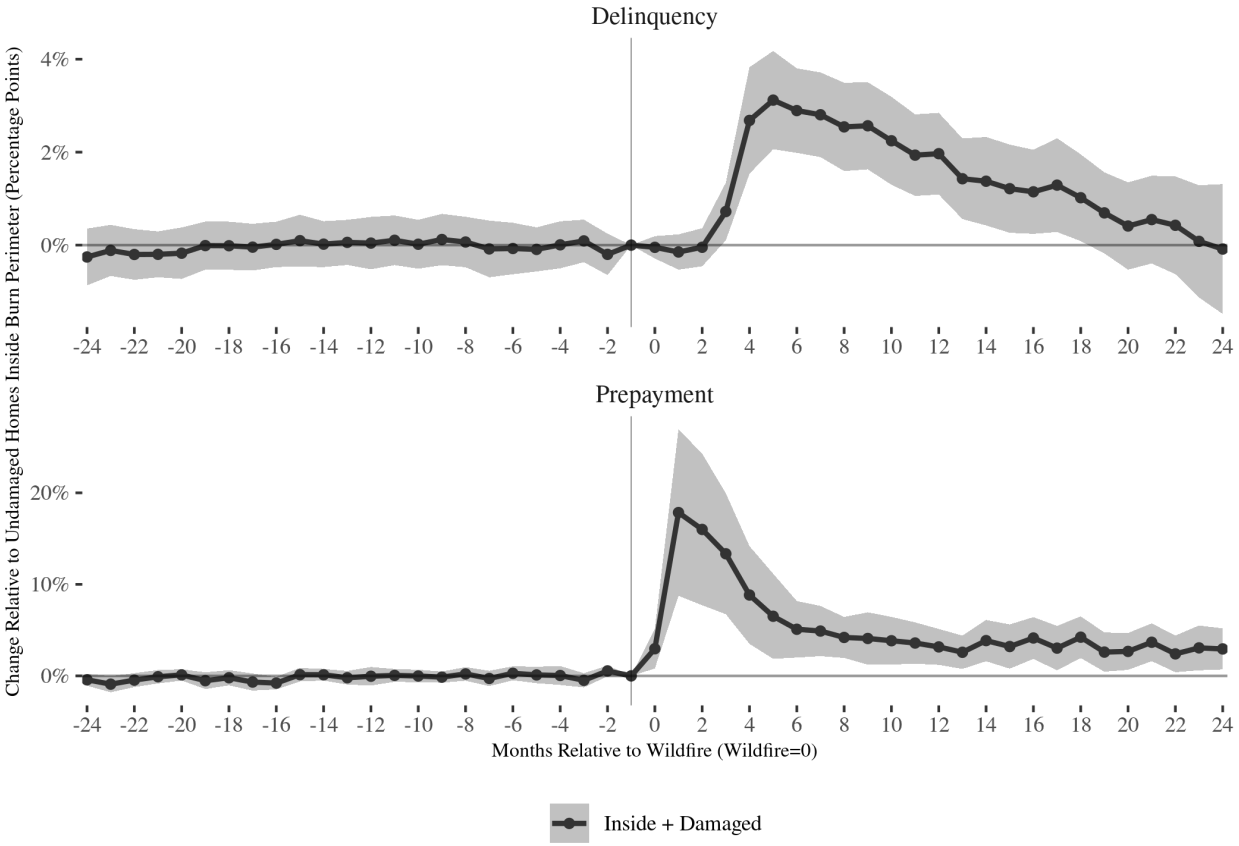

Note: Figure illustrates the coefficients for destroyed ($\beta_{1t}$) and undamaged ($\beta_{2t}$) parcels estimated from the event study specification in Equation 1. The top panel presents results for delinquency and the bottom panel presents results for prepayment. Regression controls for fire-month and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.

Figure A.7: Robustness to Excluding Outside Fire as Control Group



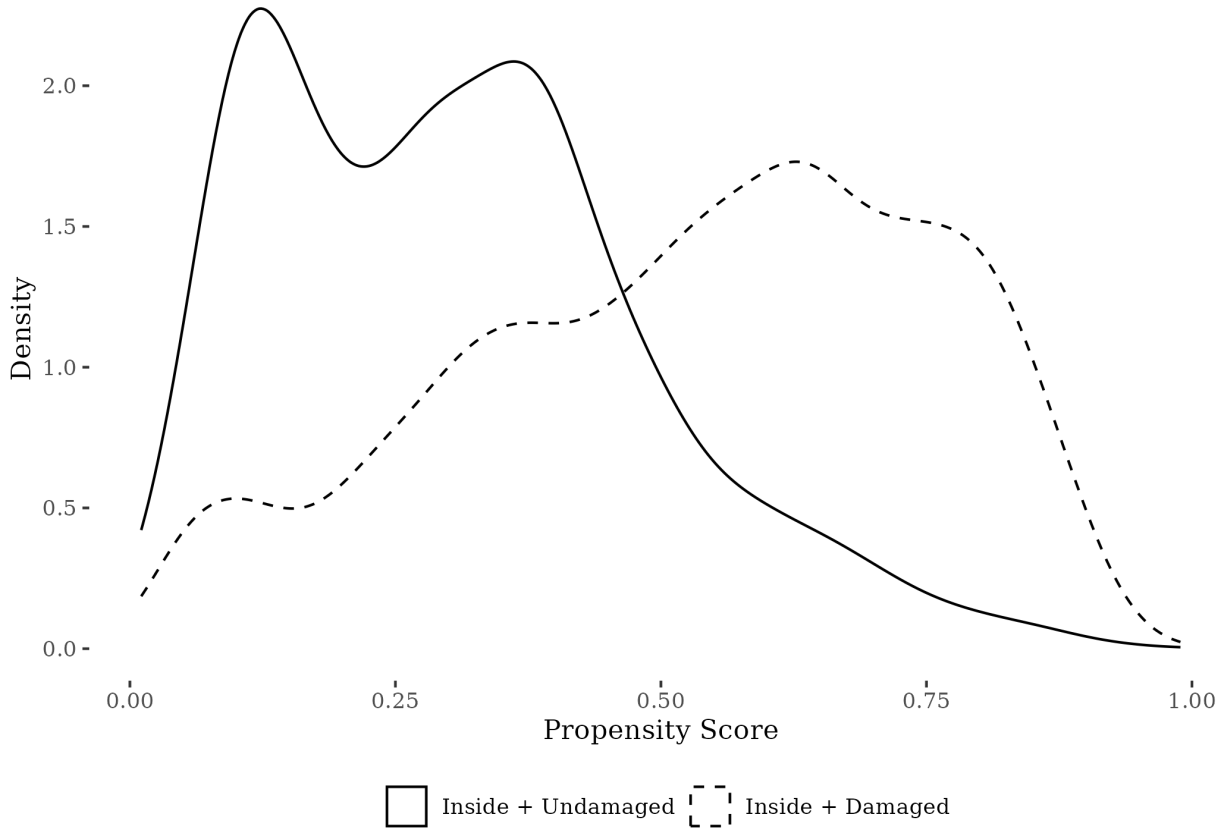

Note: Figure illustrates the event study coefficients for damaged parcels relative to undamaged parcels inside the burn perimeter. The top panel presents results for delinquency and the bottom panel presents results for prepayment. Regression controls for fire-month and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.

Figure A.8: Robustness to Excluding Outside Fire as Control Group, IPW Estimates



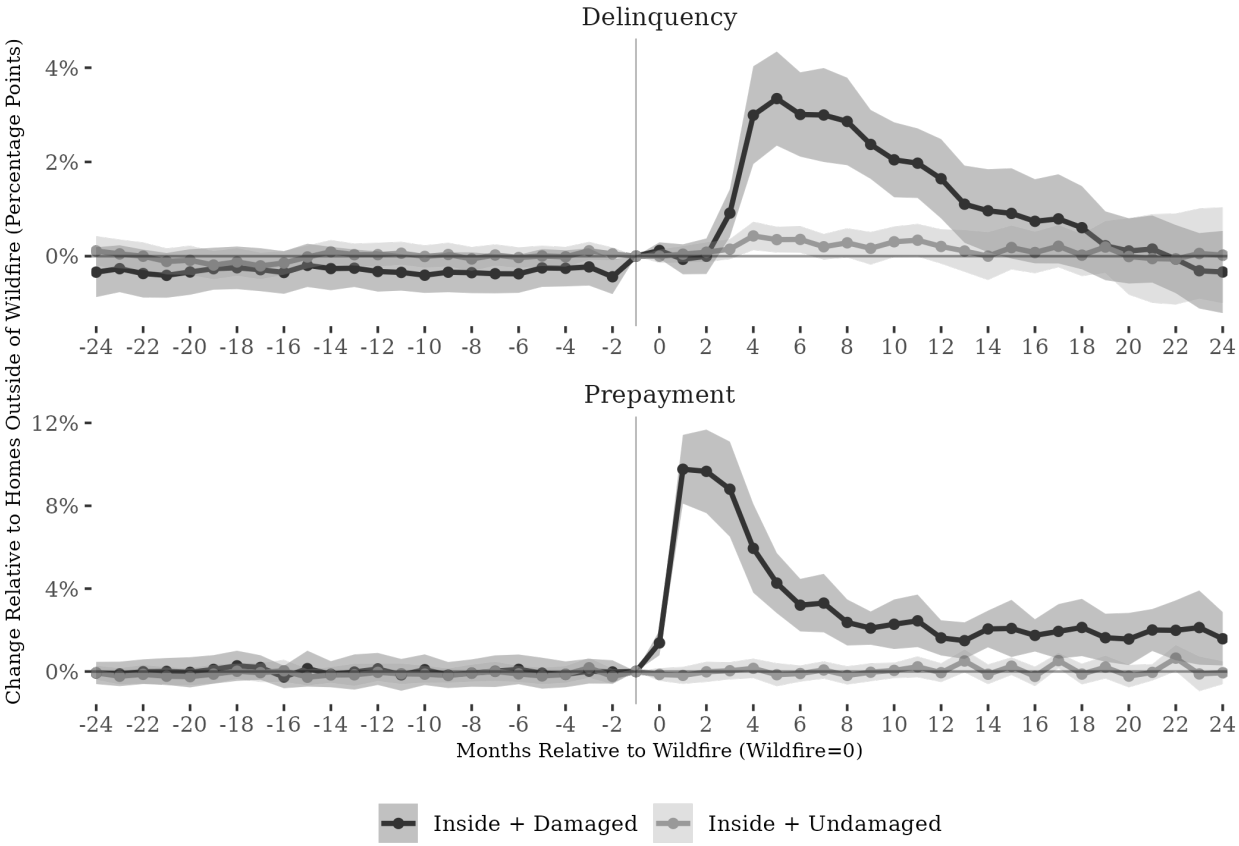

Note: Figure illustrates the inverse propensity weighted event study coefficients for damaged parcels relative to undamaged parcels inside the burn perimeter. The top panel presents results for delinquency and the bottom panel presents results for prepayment. Regression controls for fire-month and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.

Figure A.9: Propensity Score Distribution



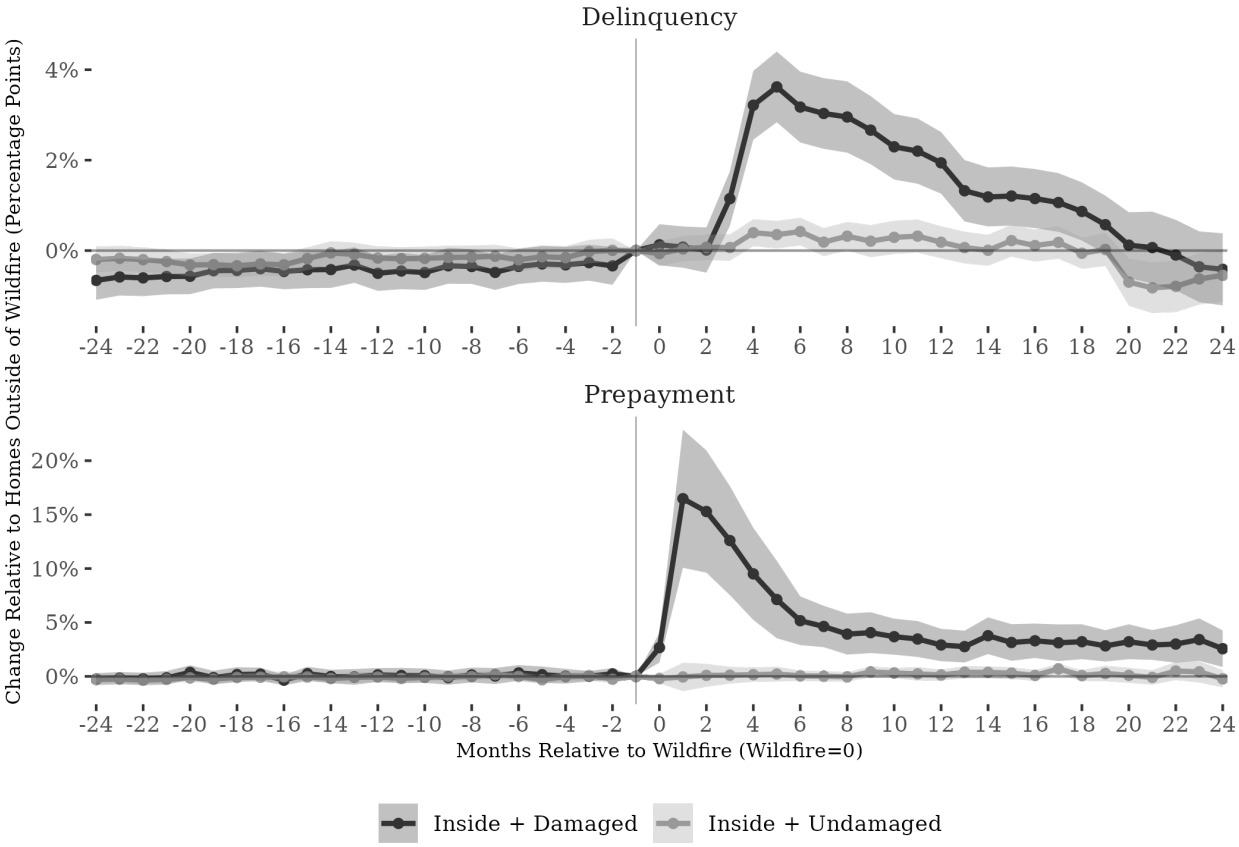

Note: Figure shows densities of propensity scores for damaged and undamaged parcels inside the burn perimeter.

Figure A.10: Robustness to Omitting Camp Fire



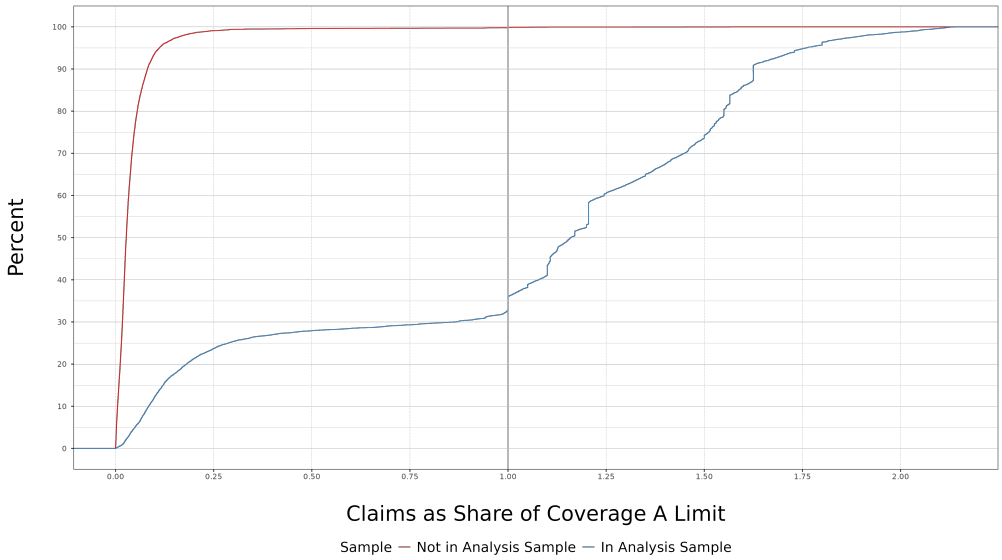

Note: Figure illustrates the coefficients for damaged ($\beta_{1t}$) and undamaged ($\beta_{2t}$) parcels estimated from the event study specification in Equation 1 after excluding the Camp Fire in 2018. The top panel presents results for delinquency and the bottom panel presents results for prepayment. Regression controls for fire-month and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.

Figure A.11: Robustness to Choice of Control Group: Properties 5 to 6 Miles Outside Fire Perimeter

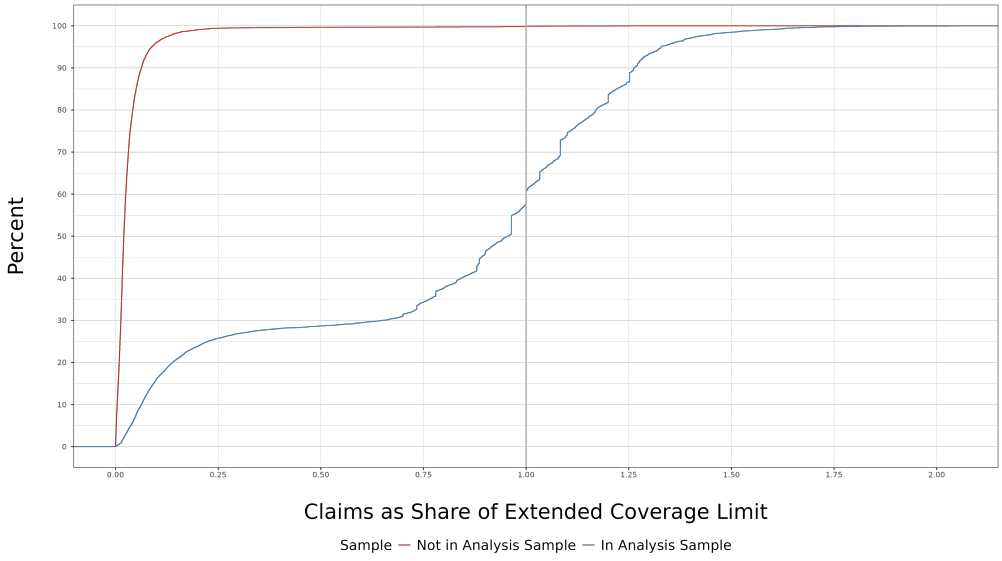



Note: Figure illustrates the coefficients for damaged ($\beta_{1t}$) and undamaged ($\beta_{2t}$) parcels estimated from the event study specification in Equation 1 using homes in the 5- to 6-mile ring outside the fire perimeter as the control group. The top panel presents results for delinquency and the bottom panel presents results for prepayment. Regression controls for fire-month and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.

Figure A.12: Distribution of Claims Relative to Coverage



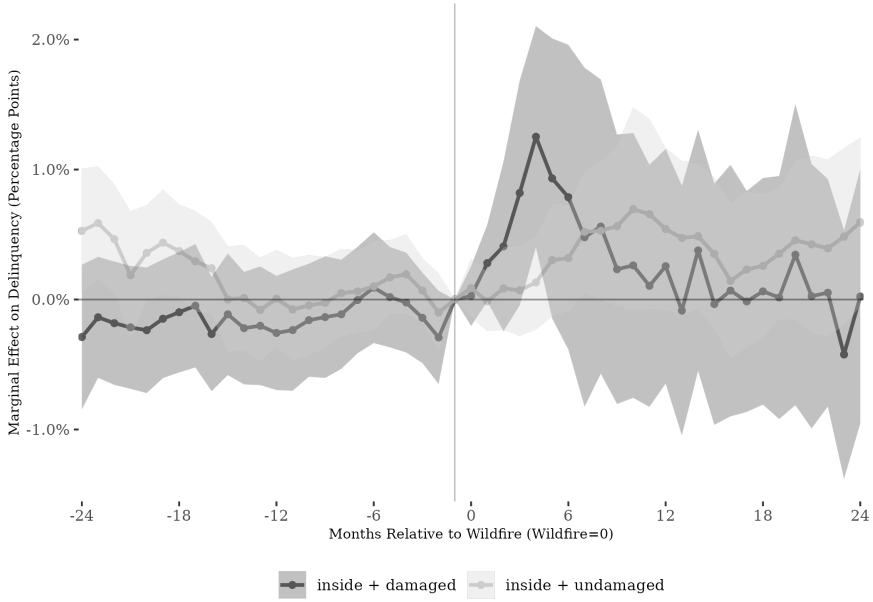

(a) Relative to Coverage A Limit



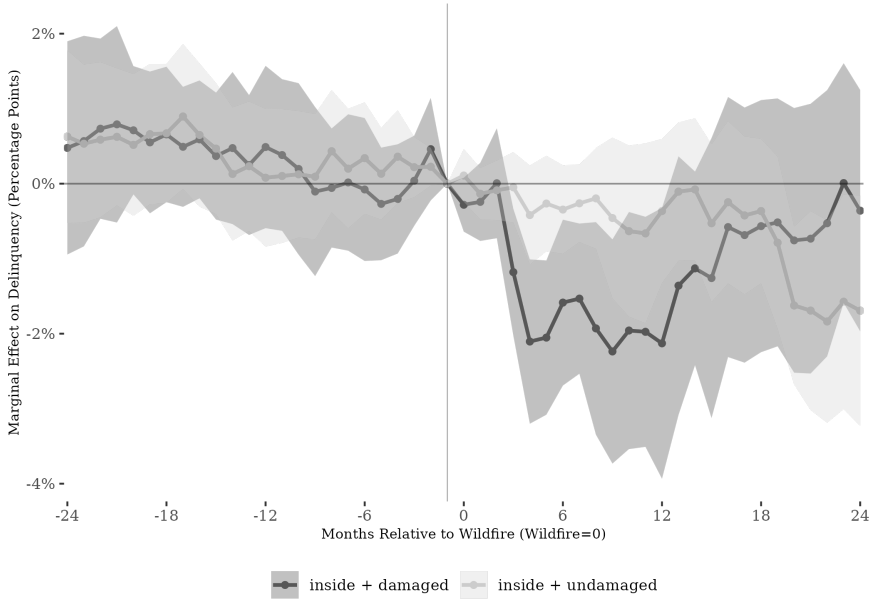

(b) Relative to ERCV Limit

Note: Figure shows the cumulative distribution of claims to coverage ratio. Panel (a) uses the coverage A limit as the denominator and Panel (b) uses the extended replacement cost (ERCV) coverage limit as the denominator. The "In Analysis Sample" distribution refers to the largest $N_{jt}$ claims in zip code $j$ and month $t$, where $N_{jt}$ is the number of observed wildfire-destroyed properties. This is the sample of claims used to calculate reimbursement gaps. The "Not in Analysis Sample" captures all other fire-related catastrophic claims.

Figure A.13: Differences in Delinquency by Interest Rate and Credit Score




(a) Interest Rate × Fire Damage




(b) Credit Score × Fire Damage

Note: Figure illustrates the coefficients for $Z_{ift}$ interacted with damaged ($\gamma_{1t}$) and undamaged ($\gamma_{2t}$) parcels estimated from the event study specification in Equation 6. Both plots present the differential impact on delinquency after the fire for loans with different interest rates (top panel) and borrowers with different credit scores (bottom panel). Coefficients estimate the marginal effect of a 1 standard deviation change in $Z_{ift}$. Regression controls for fire-month and loan fixed effects. Shaded regions indicate 95% confidence intervals clustered by zip code.

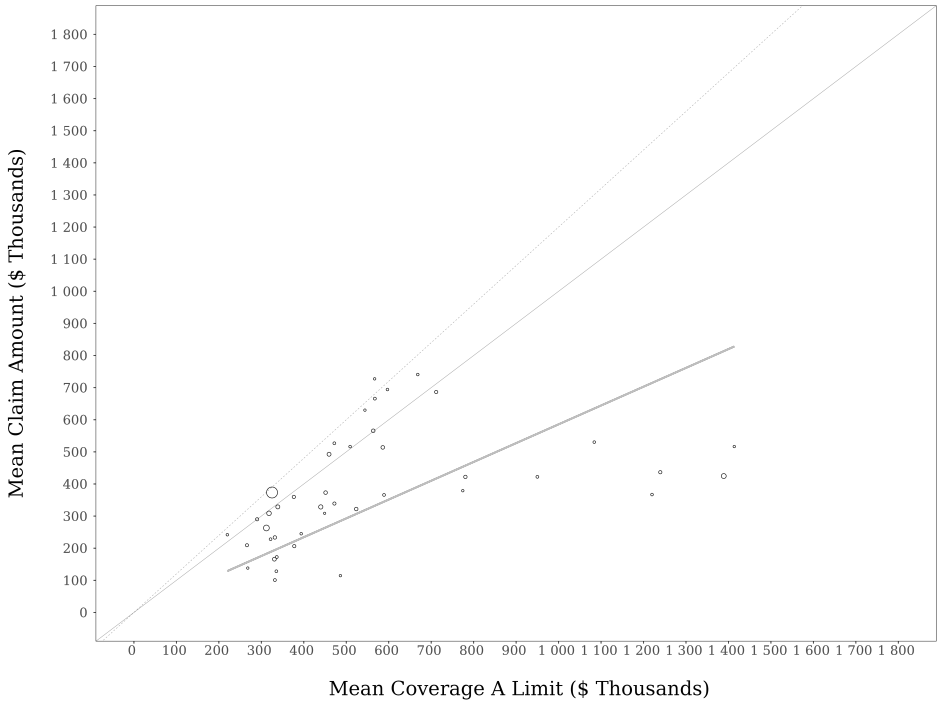



Figure A.14: Zip-Code Level Coverage and Claims

Note: Each circle represents a zip code with the size of the circle varying by the number of claims. The darker solid gray line plots the correlation between zip code mean coverage limit and mean claim amount. The solid black 45-degree line represents the counterfactual, in which claim settlements equal the Coverage A limit. The dotted line represents the counterfactual of 20 percent extended replacement cost value coverage (ERCV), in which claims equal $1.20 per dollar of Coverage A. Over 80 percent of policyholders hold at least 20 percent ERCV coverage. We include the largest $N_{jt}$ claims in zip code $j$ and month $t$, where $N_{jt}$ is the number of observed wildfire damaged-properties. Standard errors are clustered by zip code.

## A.4 Additional Tables

Table A.1: Comparing HMDA and FR Y-14M Loan Originations in California

| Variable | HMDA | Y-14 |
|---|---|---|
| $N$ Loans | 3176731 | 692040 |
| LTV | 66.28 | 65.04 |
| Loan Amount | 465988.80 | 622901.80 |
| Interest Rate | 3.70 | 3.79 |
| Debt-to-Income Ratio | | |
| <20% | 6.38 | 5.47 |
| 20% - 36% | 36.67 | 38.35 |
| 36% - 50% | 52.41 | 53.17 |
| >= 50% | 4.54 | 3.01 |

Note: Table presents mean characteristics of loans originated in California reported in two data sources: Home Mortgage Disclosure Act (HMDA) and FR Y-14M. As DTI is reported in bins in HMDA, we provide the distribution of loans across four DTI bins.

Table A.2: Reimbursement Gaps, Insured only

| | All | 2018 | 2019 | 2020 |
|---|---|---|---|---|
| Average Gap (share of limit) | 0.023 | 0.003 | 0.108 | 0.101 |
| Average Gap ($) | 9299 | 986 | 59435 | 50114 |
| Any Gap (share) | 0.248 | 0.228 | 0.467 | 0.344 |
| Average Gap, if any gap (share of limit) | 0.737 | 0.764 | 0.588 | 0.657 |
| Average Gap, if any gap ($) | 430105 | 439286 | 322397 | 403536 |
| Number of Claims | 8945 | 7440 | 45 | 1460 |
| Number of Zip-Months | 78 | 23 | 10 | 45 |

Note: Table reports reimbursement gaps. A claim is considered underpaid if the settlement is below the Coverage A limit. The deficit is the difference between the Coverage A limit and claim amount. Deficits that are represented as shares are dollar weighted by the coverage limit. We include the largest $N_{jt}$ claims in zip code $j$ and month $t$, where $N_{jt}$ is the number of observed wildfire-damaged properties, to identify the set of claims associated with wildfire damage.

Table A.3: Reimbursement Gaps, including GRCV Policies and Outliers

| Include GRCV policies | All | 2018 | 2019 | 2020 |
|---|---|---|---|---|
| Average Gap (share of limit) | 0.152 | 0.126 | 0.191 | 0.247 |
| Average Gap ($) | 64191 | 50533 | 112483 | 126340 |
| Any Gap (share) | 0.328 | 0.303 | 0.491 | 0.438 |
| Average Gap, if any gap (share of limit) | 0.789 | 0.809 | 0.640 | 0.734 |
| Average Gap, if any gap ($) | 456865 | 458171 | 400424 | 454585 |
| Number of Claims | 10268 | 8408 | 55 | 1805 |
| Number of Zip-Months | 96 | 28 | 11 | 57 |

| Include GRCV policies & outliers | All | 2018 | 2019 | 2020 |
|---|---|---|---|---|
| Average Gap (share of limit) | 0.028 | 0.012 | -0.043 | 0.086 |
| Average Gap ($) | 12631 | 5154 | -31306 | 47874 |
| Any Gap (share) | 0.301 | 0.280 | 0.370 | 0.393 |
| Average Gap, if any gap (share of limit) | 0.780 | 0.801 | 0.640 | 0.724 |
| Average Gap, if any gap ($) | 457545 | 459114 | 400424 | 454443 |
| Number of Claims | 11225 | 9125 | 73 | 2027 |
| Number of Zip-Months | 97 | 28 | 11 | 58 |

Note: Table reports reimbursement gaps, which is calculated as the amount by which the settlement is below the Coverage A limit. Gaps that are represented as shares are dollar weighted by the coverage limit. We include the largest $N_{jt}$ claims in zip code $j$ and month $t$, where $N_{jt}$ is the number of observed wildfire-destroyed properties, to identify the set of claims associated with wildfire damage. Dollar amounts are deflated to 2020 USD.

Table A.4: Correlation of Claims and Limits Within Zip Codes

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Coverage A ($) | 0.5218** | 0.2505* | 0.0328 | 0.0316 |
| | (0.1798) | (0.1062) | (0.0681) | (0.0678) |
| ERCV (Additional $) | -0.1501* | 0.0136 | -0.0336 | -0.0396 |
| | (0.0609) | (0.0874) | (0.0234) | (0.0218) |
| Policy Premium | | 628.1*** | | -79.31** |
| | | (130.6) | | (28.19) |
| Month-Year | No | No | Yes | Yes |
| Zip Code | No | No | Yes | Yes |
| Policy Type | No | No | Yes | Yes |
| Observations | 10,268 | 10,268 | 10,246 | 10,246 |

Note: Table reports OLS and fixed-effects estimates of $\hat{\beta}$ from Equation 7. ERCV (Additional $) represents the additional dollar amount extended replacement cost value coverage provides to the specific policy beyond the Coverage A limit. Policy premium measures the annual price per $100,000 of Coverage A limit. The last two columns includes fixed effects for the month and year of loss, zip code, and policy type (homeowners or dwelling fire). We include the largest $N_{jt}$ claims in zip code $j$ and month $t$, where $N_{jt}$ is the number of observed wildfire-damaged properties. Standard errors are clustered at zip code.

Table A.5: Impacts on Household Equity, including GRCV Policies

| Include GRCV policies | | Shortfall Relative to: | |
|---|---|---|---|
| | $N$ | Coverage A ($) | ERCV ($) |
| All | 9,110 | -151,317 | -232,944 |
| 2018 | 8,195 | -137,771 | -215,254 |
| 2019 | 51 | -190,188 | -260,806 |
| 2020 | 864 | -277,500 | -399,089 |
| Camp Fire Only | 6,886 | -71,773 | -122,377 |
| Excluding Camp Fire | 2,224 | -397,602 | -575,287 |

| Include GRCV policies & outliers | | Shortfall Relative to: | |
|---|---|---|---|
| | $N$ | Coverage A ($) | ERCV ($) |
| All | 9,997 | -138,690 | -215,950 |
| 2018 | 8,904 | -127,587 | -200,885 |
| 2019 | 66 | -146,964 | -214,647 |
| 2020 | 1,027 | -234,424 | -346,650 |
| Camp Fire Only | 7,351 | -67,233 | -114,659 |
| Excluding Camp Fire | 2,646 | -337,209 | -497,352 |

Note: Shortfall amounts are calculated from Equation 4. We assume the counterfactual "full" payout is capped by either the Coverage A limit or the ERCV. Each row provides the same calculation for different subsamples of our claims data. We include the largest $N_{jt}$ claims in zip code $j$ and month $t$, where $N_{jt}$ is the number of observed wildfire-destroyed properties.

Table A.6: Impacts on Household Equity, Insured Only

| | | Shortfall Relative to: | |
|---|---|---|---|
| | $N$ | Coverage A ($) | ERCV ($) |
| All | 8,013 | -107,840 | -176,755 |
| 2018 | 7,290 | -99,915 | -166,837 |
| 2019 | 43 | -147,903 | -201,218 |
| 2020 | 680 | -190,266 | -281,536 |
| Camp Fire Only | 6,215 | -52,253 | -96,855 |
| Excluding Camp Fire | 1,798 | -299,983 | -452,937 |

Note: Shortfall amounts are calculated from Equation 4. We assume the counterfactual "full" payout is capped by either the Coverage A limit or the ERCV. Each row provides the same calculation for different subsamples of our claims data. We include the largest $N_{jt}^{Ins}$ claims in zip code $j$ and month $t$, where $N_{jt}^{Ins}$ is the number of insured wildfire-destroyed properties.